

A PROOF-THEORETIC BOUND EXTRACTION THEOREM FOR MONOTONE OPERATORS IN BANACH SPACES

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ABSTRACT. We utilize a proof-theoretically tame approach to the dual of an abstract Banach space in systems amenable to methods from proof mining, as recently introduced by the author, to provide similar such systems with accompanying logical metatheorems on the extraction of uniform quantitative information from proofs pertaining to the theory of monotone set-valued operators on Banach spaces as introduced by Browder. With that, we finally extend proof mining methods to this important class of objects which are at the heart of many seminal results from nonlinear functional analysis, and the metatheorems presented here in particular provide the first logical basis for a range of recent applications of proof mining methods to this branch of mathematics. Further, we provide a characterization of the extensionality principle for these operators using the analytical notion of maximality, extending previous analogous results for accretive operators on Banach spaces and monotone operators on Hilbert spaces, and with that further illustrate the central importance and special position of extensionality issues in proof mining applications dealing with set-valued operators.

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1. INTRODUCTION

The quest for describing the computational content and strength of a mathematical theorem and developing tools to exhibit such content has been and is one of the main driving interests of proof theory. The program of proof mining, which emerged as a subfield of mathematical logic in the 1990s through the work of Ulrich Kohlenbach and his collaborators (rooting in Georg Kreisel’s program of “unwinding of proofs”, see in particular [56, 57]), aims at answering exactly this question and so is concerned with exhibiting the computational content of a mathematical theorem by analyzing its proofs as they are found in the mainstream mathematical literature using proof-theoretic tools. As such proofs are *prima facie* noneffective, e.g. involving full use of classical logic as well as many non-computational principles, this is a highly non-trivial task. However, relying on a comprehensive theoretical underpinning developed using a wide range of methods from proof theory like functional interpretations and majorizability, this research program has been very successful in a wide variety of areas of mathematics, and even more noticeably so in the areas of nonlinear analysis and optimization. We refer to [40] for the comprehensive monograph on proof mining and its applications until 2008 and we refer to the survey article [52] for details on the earlier development of proof mining as well as the more recent survey articles [41, 43] for more current developments.

While the case studies of this research program are presented using the usual mathematical means of the area of application in question and without any apparent use of logical methods, they nevertheless all rely on central logical results explaining and governing these extractions. These substrata of proof mining, often called *logical metatheorems* or *bound extraction theorems*,

utilize a composition of various proof-theoretic devices like negative translations (see e.g. [59]), Kreisel’s *modified realizability* [58] or Gödel’s *functional (Dialectica) interpretation* [27] and their variants to provide both a logical system and a general theorem, the so-called logical metatheorem, about that system so that

- (1) the system tethered to the metatheorem is suitably designed so that it allows for the formalization of large classes of objects and proofs from the core mathematical literature of the intended area of application;
- (2) the associated logical metatheorem guarantees that for large classes of proofs (e.g. potentially involving a wide range of non-computational ‘ideal principles’ and using classical logic) carried out in this system, one can extract effective, tame and highly uniform computational information for the theorem proved thereby.

Even further, the proof of the logical metatheorems provides an algorithm to (in principle) extract this information from a given proof. Also, the complexity of this extracted information will faithfully represent the logical strength of the key principles used in the (formal) proof.

On the basis of these metatheorems, the research program of proof mining has lead to hundreds of new results in the respective areas of application over the last three decades, in particular substantiating item (1) above that the systems designed in the context of such metatheorems really are practically usable. Examples of such metatheorems can in particular be found in [24, 26, 28, 39, 40, 51, 60, 62, 72, 85, 80, 76, 77, 92].

As mentioned before, one main focus for the applications in the proof mining program in the recent years has been on proofs from nonlinear analysis and optimization, in particular those that utilize so-called monotone set-valued operators in Hilbert spaces. These operators go back to the fundamental work of Minty [70, 71] and are today one of the main cornerstones of a modern operator-theoretic approach to convex analysis on Hilbert spaces as is particularly well illustrated by the seminal monograph of Bauschke and Combettes [7]. Notable examples of proof mining case studies concerned with these operators in particular are [42],¹ providing a low-complexity and highly uniform rate of asymptotic regularity in the context of Bauschke’s solution [3] to the zero displacement conjecture, and the many case studies on (variants of) the influential proximal point algorithm (going back to the seminal work of Martinet [69] and Rockafellar [89]) as e.g. treated in [21, 22, 45, 46, 47, 50, 63, 64, 73], among many others. If one moves down from Hilbert to Banach spaces, the notion of a monotone operator generalizes in distinct ways based on the multiple variants of equivalently phrasing monotonicity in the context of Hilbert spaces. One such generalization leads to the notion of accretive operators which go back to the seminal work of Kato [34] and are central e.g. to the study of differential equations and semigroup theory (with a range of proof mining case studies dedicated to this class of operators already carried out, where we in particular want to mention the works [25, 48, 54, 74, 79] besides some of the papers on the proximal point algorithm mentioned before that are actually phrased in terms of these accretive operators). The other central avenue for generalizing the concept of monotonicity leads to the notion of a monotone operator in Banach spaces in the sense of Browder [14, 15] which are central e.g. to many parts of convex analysis over Banach spaces.

While the accretive operators on Banach spaces and the monotone operators on Hilbert spaces have previously received a logical treatment in terms of associated metatheorems in the style of proof mining in [76], thereby providing the first logical footing for all of the previously mentioned case studies, the monotone operators on Banach spaces in the sense of Browder have

¹We refer to [53] for a discussion of the logical details of this proof mining application and to [93, 94] for further extensions of that work.

so far remained elusive from such a treatment, which is in particular due to the heavy use made in their theory of the dual of the associated Banach space, an object that itself evaded any type of proof-theoretically tame² treatment in the style of proof mining until it was recently treated in [77].

It is now the aim of this paper to provide such a treatment in the style of proof mining for the monotone operators on Banach spaces in the sense of Browder by defining associated formal systems and proving a corresponding logical metatheorem, and in that way paving the way for many further applications of methods from proof mining to this central area of modern convex analysis on Banach spaces. This is achieved by carefully combining and extending the ideas of the systems presented in [76] for the treatment of monotone operators in Hilbert spaces with those presented in [77] for the treatment of the dual of a Banach space. In that way, the present paper also further elucidates the naturalness and applicability of the methods developed in the above mentioned papers to treat set-valued operators of various types as well as to treat the dual of a Banach space and that these approaches are indeed as flexible as hoped for.

Besides treating these operators, this paper also provides a proof-theoretic treatment of other central surrounding objects that are crucial for developing their theory, like in particular the so-called resolvents of such operators defined relative to a convex function f as first considered in the seminal work of Eckstein [23] in the context of finite-dimensional spaces and then highlighted as a central tool of monotone operators in general Banach spaces in particular in the work of Bauschke, Borwein and Combettes [6]. These resolvents can then in particular be used for a characterization of maximally monotone operators similar as in Minty's theorem which will be crucial for the systems devised in this paper for these operator. Further, they also are the essential objects for many central applications of these operators to convex analysis like in approaches to iterative methods used in mathematical programming. In that way, to illustrate the usability of the defined systems for these objects, we show that the main properties of the operator and the associated relativized resolvents are provable therein. This in turn relies on first establishing further results from the duality theory of convex functions, in particular involving the Fenchel conjugate f^* of f , in the respective system, extending [77] and also being potentially of use in other contexts.

Through these considerations, we arrive at a practically suitable system $\mathcal{B}^\omega$ for treating convex functions, their conjugates and gradients as well as monotone operators with total relativized resolvents, for which we establish an associated logical metatheorem. A special case of this metatheorem roughly takes the following form: If a sentence of the form $\forall z^\tau \exists v^0 G_\exists(z, v)$ is provable in $\mathcal{B}^\omega$, where τ is a suitable finite type that is “low enough” and $G_\exists$ is an existential formula, then from the corresponding proof one can extract a computable functional Φ which bounds v in terms of z in a uniform manner, that is

$$\forall z \forall z^* \succsim_\tau z \exists v \leq_0 \Phi(z^*) G_\exists(z, v),$$

where $z^* \succsim_\tau z$ represents that z^* provides a bound on z in the sense of an extended version of Howard's majorization [30] (an Bezem's refinement [9], see in particular Definition 5.5 later). This bound is validated in a standard set-theoretic model $\mathcal{S}^{\omega, X, X^*}$ based on a maximally monotone operator A with $A^{-1}0 \neq \emptyset$ and a convex, supercoercive and Fréchet differentiable function $f : X \rightarrow \mathbb{R}$ where $\nabla f, \nabla f^*$ are uniformly continuous on bounded subsets, as treated in the

²The notion of *proof-theoretic tameness* is here understood in the sense of [44], i.e. referring to the fact that although an area of mathematics could be subject to well-known Gödelian phenomena, they nevertheless “seem to be tame in the sense of allowing for the extraction of bounds of rather low complexity” (as phrased in [44]). We refer also to [66, 67] for further discussions of these types of phenomenas and their implications for logic and mathematics.

system, giving rise to associated relativized resolvents of A . Next to the uniformity on z , the bound Φ is further uniform in these data that define $\mathcal{S}^{\omega, X, X^*}$, in that it only depends on a modulus of supercoercivity for f and moduli of uniform continuity for $\nabla f, \nabla f^*$ (see Section 2.2.3 later on) as well as upper bounds on the quantities $|f(0)|, \|\nabla f(0)\|, |f^*(0)|, \|\nabla f^*(0)\|$ and $\|p\|$, where $p \in A^{-1}0$.

The precise metatheorem formulated in this paper (see Theorem 5.7 later on) refines this rough structure by incorporating slightly more complicated sentences and allowing additional assumptions of a suitable form, extending $\mathcal{B}^\omega$.

Next to this main result, we also show that the equivalence between maximality of the operator and extensionality, which was one of the central theoretical results established in [76] for monotone operators on Hilbert spaces and which illustrates some theoretical limitations of such systems, extends to these new objects. We further provide similar such equivalences between the extensionality of the zero set of the operator and a strong characterization of the fixed point set of the resolvent, which retroactively also apply to [76].

The applicability of the systems defined in the present paper is then in particular illustrated from the practical perspective through the recent case studies given in [75, 81, 84],³ situated in the area of convex analysis on Banach spaces. However, we expect that these metatheorems allow for many further new case studies to be carried out in the areas discussed above and for that, we want to also in particular mention the well-known works [1, 4, 13, 19, 32], next to many others, as promising future applications since, by inspection of the proofs, they also seem to be formalizable in (suitable extensions of) the systems introduced here.

2. LOGICAL AND MATHEMATICAL PRELIMINARIES

In this section, we briefly survey both the mathematical and logical objects and results that will play a key role in developing the new formal systems of this paper.

2.1. Monotone operators on Banach spaces and relativized resolvents. We begin by discussing the (very minimal) mathematical background of the theory of monotone operators in Banach spaces as well as of related objects relevant for this paper. Further definitions will be given throughout the sections as needed and we refer to the standard works [88, 90, 99] for further results and references.

2.1.1. A primer on convex analysis. Throughout, if not specified otherwise, let X be a Banach space with norm $\|\cdot\|$ and dual space X^* with the associated dual norm (also denoted by) $\|\cdot\|$, defined via

$$\|x^*\| := \sup_{\|x\| \leq 1} \langle x, x^* \rangle,$$

for $x^* \in X^*$, where we write $\langle x, x^* \rangle$ for the function value of x^* applied to $x \in X$. We write $B_r(x)$ for the open and $\overline{B}_r(x)$ for the closed ball in X with radius $r > 0$ centered at x .

Let $f : X \rightarrow (-\infty, +\infty]$ be a given function with extended real values. In the following, we will always at least assume three properties for f :

- (1) f is proper, i.e.

$$\text{dom } f := \{x \in X \mid f(x) < +\infty\} \neq \emptyset,$$

- (2) f is lower-semicontinuous, i.e.

$$\forall x \in \text{dom } f \forall y < f(x) \exists \delta > 0 \forall z \in B_\delta(x) (f(z) > y),$$

³We also want to note the work [83] here where the formal treatment of monotone operators on Banach spaces presented here was instrumental for introducing novel analogous notions in a nonlinear setting.

(3) f is convex, i.e.

$$\forall x, y \in \text{dom} f \forall \lambda \in [0, 1] (f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)).$$

Crucial to the study of convex functions in nonlinear analysis and optimization is the use of so-called subgradients.

Definition 2.1 (Subdifferential). Let $x \in \text{intdom} f$, where $\text{intdom} f$ denotes the interior of $\text{dom} f$. We define $\partial f(x)$, the subdifferential of f at x , via

$$\partial f(x) := \{x^* \in X^* \mid f(x) + \langle y - x, x^* \rangle \leq f(y) \text{ for all } y \in X\}.$$

A subgradient of f at x , i.e. a point $x^* \in \partial f(x)$, can in many places be substituted for a gradient of the function when it is not differentiable. In this work however, the focus will be on convex functions which are actually differentiable in the following ways:

Definition 2.2 (Gâteaux and Fréchet differentiability). The function f is called Gâteaux differentiable at $x \in \text{intdom} f$ if there exists a $\nabla f(x) \in X^*$ such that

$$\langle h, \nabla f(x) \rangle = \lim_{t \rightarrow 0^+} \frac{f(x + th) - f(x)}{t}$$

for any $h \in X$. The function f is called Fréchet differentiable at $x \in \text{intdom} f$ if there exists a $\nabla f(x) \in X^*$ such that

$$\lim_{\|h\| \rightarrow 0} \frac{|f(x + h) - f(x) - \langle h, \nabla f(x) \rangle|}{\|h\|} = 0.$$

The function f is called Fréchet (or Gâteaux) differentiable on a set $D \subseteq X$ if it is Fréchet (or Gâteaux) differentiable at every point $x \in D$ and f is called uniformly Fréchet differentiable on D if the above limit is attained uniformly in $x \in D$.

Crucial for a formal approach to such derivatives later on will be an equivalent characterization of Gâteaux and Fréchet derivatives presented in the following proposition that does not rely on the use of the above limits per se.

Proposition 2.3 (see e.g. [99]). *Let $x \in \text{intdom} f$. Then, the following are equivalent:*

- (1) f is Fréchet (respectively Gâteaux) differentiable at x .
- (2) There exists a selection of ∂f that is norm-to-norm (respectively norm-to-weak) continuous at x .

Further, it holds that if f is Gâteaux differentiable at x , then $\partial f(x) = \{\nabla f(x)\}$.

Further, also the uniform Fréchet differentiability connects to a continuity property of the gradient:

Proposition 2.4 (essentially [86]). *If f is uniformly Fréchet differentiable on bounded sets and ∇f is bounded on bounded sets, then ∇f is uniformly norm-to-norm continuous on bounded sets.*

The main object for the duality theory of a convex function f in Banach spaces is the Fenchel conjugate $f^* : X^* \rightarrow (-\infty, +\infty]$, concretely defined by

$$f^*(x^*) := \sup_{x \in X} (\langle x, x^* \rangle - f(x)).$$

Crucial for a formal treatment of this derived object later on will be the following result that characterizes when f^* is bounded on bounded sets.

Proposition 2.5 (see e.g. [5]). *Call f supercoercive if*

$$\lim_{\|x\| \rightarrow +\infty} \frac{f(x)}{\|x\|} = +\infty.$$

Then, the following are equivalent:

- (1) *f is supercoercive.*
- (2) *f^* is bounded on bounded subsets.*

In particular, both imply that $\text{dom} f^ = X^*$.*

We now introduce the main class of functions considered in this paper, first in the way as it is commonly introduced in the convex analysis literature:

Definition 2.6 ([5]). A function f is called:

- (1) essentially smooth if ∂f is locally bounded and single-valued on its domain,
- (2) essentially strictly convex if $(\partial f)^{-1}$ is locally bounded and f is strictly convex on every convex subset of $\text{dom} \partial f$,
- (3) Legendre if it is both essentially smooth and essentially strictly convex.

Over reflexive spaces, the class of Legendre functions can be equivalently recognized via a particularly nice differentiability property for both f and its conjugate f^* which serves as the central definition of this class of functions in this paper.

Proposition 2.7 ([5]). *If X is reflexive, then f is Legendre if, and only if both*

- (1) *It holds that $\text{intdom} f \neq \emptyset$, that f is Gâteaux differentiable on $\text{intdom} f$, and $\text{dom} \nabla f = \text{intdom} f$.*
- (2) *It holds that $\text{intdom} f^* \neq \emptyset$, that f^* is Gâteaux differentiable on $\text{intdom} f^*$, and $\text{dom} \nabla f^* = \text{intdom} f^*$.*

In particular, if X is reflexive and f is Legendre, then the associated gradients $\nabla f : \text{intdom} f \rightarrow \text{intdom} f^*$ and $\nabla f^* : \text{intdom} f^* \rightarrow \text{intdom} f$ are bijective, with $(\nabla f)^{-1} = \nabla f^*$ (see Theorem 5.10 in [5]).

Also crucial for the study of such functions and related objects, in particular regarding analytical aspects of associated iterations, is the notion of the Bregman distance $D_f : \text{dom} f \times \text{intdom} f \rightarrow [0, +\infty)$ as introduced in the seminal work [13] and which is defined by

$$D_f(x, y) := f(x) - f(y) - \langle x - y, \nabla f(y) \rangle.$$

Dual to this distance notion is the function $W_f : \text{dom} f \times \text{dom} f^* \rightarrow [0, +\infty)$ defined by

$$W_f(x, x^*) := f(x) - \langle x, x^* \rangle + f^*(x^*).$$

For this function, we will rely later on a few further properties: If f is Legendre and if X is reflexive, one in particular has that

$$W_f(x, \nabla f(y)) = D_f(x, y)$$

for all $x \in \text{dom} f$ and $y \in \text{intdom} f$ as well as that W_f is convex in its right argument and satisfies the inequality

$$W_f(x, x^*) \leq W_f(x, x^* + y^*) - \langle \nabla f^*(x^*) - x, y^* \rangle$$

for any $x \in \text{dom} f$ and any $x^*, y^* \in \text{dom} f^*$ (see [68]).

Example 2.8. The most canonical example of a Legendre function is $f(x) := \frac{1}{2}\|x\|^2$ in a uniformly convex and uniformly smooth Banach space $(X, \|\cdot\|)$. Indeed, in this case, we have $f^*(x^*) = \frac{1}{2}\|x^*\|^2$ and both f and f^* are Fréchet differentiable with $\nabla f = J$ and $\nabla f^* = J^{-1}$, where J is the normalized duality mapping

$$J(x) := \{x^* \in X^* \mid \langle x, x^* \rangle = \|x\|^2 = \|x^*\|^2\}$$

which in this context is single-valued and bijective (see e.g. [96]). If X is a Hilbert space, as J then reduces to the identity Id (after identifying X^* with X), we get $\nabla f = \nabla f^* = \text{Id}$, and it then further holds that $D_f(x, y) = \frac{1}{2}\|x - y\|^2$.

This example in fact generalizes to functions $f(x) := \frac{1}{p}\|x\|^p$ for $p \in (1, \infty)$, and moreover the assumptions on the space X can be slightly weakened. In particular, an f as above is Legendre if, and only if, X is smooth and rotund (that is, $\|\cdot\|^2$ is strictly convex). This result, together with associated formulas for the (sub)gradients, can be found in Lemma 6.2 of [5].

Already in finite dimensions, there are much more sophisticated examples of Legendre functions f , and we refer in particular to [4] for discussions in that vein. Many more examples in infinite dimensions can also be found in [5] (see also [6]).

We refer to Example 2.20 later on for a discussion on how, for the cases surveyed in Example 2.8 above, the key moduli featuring in our metatheorem can be instantiated.

2.1.2. Monotone operators, maximality and relativized resolvents. We now introduce one of the two main objects of concern for this paper, the so-called monotone set-valued operators. Operator-theoretic monotonicity arose in the 1960's with the work of Minty [70, 71] as a crucial concept in the context of convex analysis on Hilbert spaces. Minty's notion was subsequently extended to Banach space by Browder [14, 15], where the use of the inner product in the characterizing condition was replaced by the use of functionals from the dual space, and it has since then become one of the prime notions in convex analysis on Banach spaces (recall the discussion in the introduction).

Definition 2.9 ([14, 15]). Let $A : X \rightarrow 2^{X^*}$ be a set-valued operator. The operator A is called monotone if

$$\langle x - y, x^* - y^* \rangle \geq 0$$

for all $(x, x^*), (y, y^*) \in A$.

Further, A is called maximally monotone if its graph is not strictly contained in the graph of another monotone operator.

Example 2.10. The most canonical example of a monotone operator in the sense of Definition 2.9 is the subdifferential ∂g of a convex lower-semicontinuous proper function $g : X \rightarrow (-\infty, +\infty]$ (recall Definition 2.1). In particular, this operator is maximally monotone (as already due to Rockafellar [88]), and $(\partial g)^{-1}0 = \text{argmin } g$. Operators of this form already cover many well-known examples. In particular, we have $\partial \iota_C = N_C$ for a nonempty convex set $C \subseteq X$, where ι_C denotes the characteristic function of C (see also Remark 3.3 later on) and N_C is the so-called normal cone operator for C (see [6]). Already in the setting of Hilbert spaces, various other operators from practice can be found in [7].

Monotone operators are generally studied through the use of a certain derived object called the resolvent which in the case of a set-valued monotone operator $A : X \rightarrow 2^X$ on a Hilbert space X commonly takes the form of

$$\text{Res}_A := (\text{Id} + A)^{-1}.$$

In the context of monotone operators $A : X \rightarrow 2^{X^*}$ on Banach spaces X , where the image space makes crucial use of the dual X^* of X , one is lead to replacing the use of the identity in the above definition by a suitable function mapping from X to X^* . In the context of smooth Banach spaces, such a function is often taken to be the (in this context single-valued) normalized duality map J already discussed in Example 2.8 above, with the resolvent then defined as

$$\text{Res}_A := (J + A)^{-1} \circ J.$$

These resolvents occur for example in [33, 55] (see also [6, 8]). Abstracting from this concrete choice, one can consider a notion where J is replaced by the gradient ∇f of a (mostly) general convex function f . This even has impact already in finite-dimensional spaces where such resolvents were originally considered by Eckstein [23] as the gradient of a suitably chosen function f may ease the computations required to evaluate the resolvent. Subsequently, these relativized resolvents were also studied by Bauschke, Borwein and Combettes [6] under the name of “D-resolvents” and Reich and Sabach [87] under the name of “resolvents of A relative to f ”.

Concretely, the f -resolvents of A (as we will also call them in this paper) are now defined as follows.

Definition 2.11 ([6, 23]). Let $A : X \rightarrow 2^{X^*}$ be a set-valued operator. If f is Gâteaux differentiable on $\text{intdom} f$, the f -resolvent $\text{Res}_A^f : X \rightarrow 2^X$ of A is the operator defined by

$$\text{Res}_A^f(x) := ((\nabla f + A)^{-1} \circ \nabla f)(x)$$

for $x \in \text{intdom} f$ and by $\text{Res}_A^f(x) := \emptyset$ otherwise.

Example 2.12. For specific functions f , the resolvent Res_A^f returns familiar objects. In the most canonical case $f(x) := \frac{1}{2} \|x\|^2$ over a uniformly convex and uniformly smooth Banach space X , we have $\nabla f = J$ (recall Example 2.8), so that we obtain $\text{Res}_A^f = (J + A)^{-1} \circ J$ in that case, as motivated already before. If X is a Hilbert spaces, then $\nabla f = \text{Id}$ (recall again Example 2.8), so that we obtain $\text{Res}_A^f = (\text{Id} + A)^{-1}$, which is simply the usual resolvent of a monotone operator in Hilbert spaces (see e.g. [7]).

Moving to specific operators A , for the case $A := \partial g$ of a subdifferential for a convex lower-semicontinuous proper function g (recall Definition 2.1 and Example 2.10), the associated f -resolvent just reduces to the f -proximal map of g , that is

$$\text{Res}_{\partial g}^f(x) = \text{argmin}_{y \in \text{dom} f} \{g(y) + D_f(y, x)\}.$$

This map, which is sometimes also called the “D-prox operator” (see e.g. [6]), goes back to the work of Censor and Zenios [20] (see also [23]), and represents a variant of the usual proximal map, modulated by the Bregman distance instead of the squared distance. Indeed, if X is a Hilbert space and $f(x) := \frac{1}{2} \|x\|^2$, since $D_f(x, y) = \frac{1}{2} \|x - y\|^2$ in that case (recall Example 2.8), we in particular obtain

$$\text{Res}_{\partial g}^f(x) = \text{argmin}_{y \in X} \left\{ g(y) + \frac{1}{2} \|x - y\|^2 \right\},$$

which is just the usual proximal map of g (see e.g. [7]). If we consider the normal cone $N_C = \partial \iota_C$ for a nonempty convex set $C \subseteq X$ (recall again Example 2.10), the associated f -resolvent reduces to the Bregman projection w.r.t. f [13] (see also e.g. [5, 6]), that is

$$\text{Res}_{N_C}^f(x) = \text{argmin}_{y \in C \cap \text{dom} f} D_f(y, x),$$

and hence to the usual metric projection in Hilbert spaces with $f(x) := \frac{1}{2} \|x\|^2$. We refer to the references mentioned in Examples 2.8 and 2.10 for further examples of such f -resolvents.

The following properties are essential for the resolvent relative to f :

Proposition 2.13 ([6]). *Let f be Gâteaux differentiable and strictly convex on $\text{intdom} f$ and let A be a monotone operator such that $\text{intdom} f \cap \text{dom} A \neq \emptyset$. Then the following statements hold:*

- (1) $\text{dom}(\text{Res}_A^f) \subseteq \text{intdom} f$ and $\text{ran}(\text{Res}_A^f) \subseteq \text{intdom} f$,
- (2) Res_A^f is single-valued on its domain,
- (3) $\text{Fix}(\text{Res}_A^f) = \text{intdom} f \cap A^{-1}0$ where $\text{Fix}(\text{Res}_A^f)$ is the set of fixed points of Res_A^f ,
- (4) Res_A^f is Bregman firmly nonexpansive on its domain, i.e. for any $x, y \in \text{dom}(\text{Res}_A^f)$:
$$\langle \text{Res}_A^f x - \text{Res}_A^f y, \nabla f \text{Res}_A^f x - \nabla f \text{Res}_A^f y \rangle \leq \langle \text{Res}_A^f x - \text{Res}_A^f y, \nabla f x - \nabla f y \rangle.$$

Further, the classical result for monotone operators in Hilbert spaces established by Minty [71] that maximal monotonicity is equivalent to the totality of the resolvents extends to these resolvents relative to f under suitable assumptions on f :

Proposition 2.14 ([8]). *Let X be reflexive. Let A be monotone and assume that $f : X \rightarrow \mathbb{R}$ is Gâteaux differentiable, strictly convex and cofinite (i.e. $\text{dom } f^* = X^*$). Then A is maximally monotone if and only if $\text{ran}(A + \nabla f) = X^*$.*

Important for the study of resolvents are their corresponding Yosida approximates defined by

$$A_\gamma^f(x) := \frac{1}{\gamma} \left(\nabla f(x) - \nabla f \text{Res}_{\gamma A}^f(x) \right)$$

for a given $\gamma > 0$. It follows essentially by the definitions of $\text{Res}_{\gamma A}^f$ and A_γ^f (see e.g. [87]) that

$$(\text{Res}_{\gamma A}^f x, A_\gamma^f x) \in A$$

for any $\gamma > 0$ and any $x \in \text{dom}(\text{Res}_{\gamma A}^f)$.

2.2. A formal system for a normed space and its dual as well as uniformly Fréchet differentiable functions and conjugates. We now move to the logical preliminaries and in this subsection discuss the central systems for proof mining relevant for the paper. The central system for all of these considerations will be the system $\mathcal{D}^\omega$ as introduced in [77] that allows for a simultaneous treatment of a normed space and its dual, as well as its extensions as also discussed in [77] which allow for the treatment of a convex function together with its conjugate and the associated Fréchet derivatives. In the following, we therefore sketch the key aspects of these systems together with some of the other underlying formal background.

2.2.1. A system for classical analysis in all finite types. At first, as is the case for essentially all other modern systems of proof mining, also the system $\mathcal{D}^\omega$ and its extensions rely on a suitably strong system $\mathcal{A}^\omega = \text{WE-PA}^\omega + \text{QF-AC} + \text{DC}$ for classical analysis in all finite types. In the following, we first give a brief overview of this system and its main features relevant to this paper but otherwise refer to [26, 39, 40, 97] for any further details. Further, we in particular follow [40] in regards to our conventions for denoting types.

At first, the underlying set of types T is defined recursively via

$$0 \in T, \quad \xi, \tau \in T \Rightarrow \tau(\xi) \in T$$

and the finite types are stratified by their degrees defined by

$$\deg(0) := 0, \quad \deg(\tau(\xi)) := \max\{\deg(\tau), \deg(\xi) + 1\}.$$

Pure types, i.e. types τ that either are 0 or are of the form $\tau = 0(\rho)$ where ρ is another pure type, are abbreviated as usual by natural numbers via recursively defining $n + 1 := 0(n)$.

The language of the finite type systems $\text{WE-PA}^\omega/\mathcal{A}^\omega$ is then a many-sorted language, with variables and quantifiers for all finite types, that is extended with constants 0 for zero, S for successor, $\Sigma_{\xi,\tau}, \Pi_{\delta,\xi,\tau}$ for the so-called *combinators* (which originate from Schönfinkel [91] as well as Curry and Howard, see e.g. [31] for the latter) as well as constants $\underline{R}_\xi = (R_1)_\xi, \dots, (R_k)_\xi$ for simultaneous primitive recursion in the sense of Gödel [27] and Hilbert [29] (see also [40]) for tuples of types ξ .

The only relation symbol is $=_0$ for equality at type 0 and new terms are formed only via application: if t is a term of type $\tau(\xi)$ and s a term of type ξ , then $t(s)$ is a term of type τ . Higher type equality is treated as a defined notion via

$$s =_\xi t := \forall y_1^{\xi_1}, \dots, y_k^{\xi_k} (sy_1 \dots y_k =_0 ty_1 \dots y_k)$$

for terms s, t of type $\xi = 0(\xi_k) \dots (\xi_1)$.

The system WE-PA^ω then arises from a usual finite-type variant of Peano arithmetic (see [40, 97]) together with only the following weak *rule of quantifier-free extensionality*

$$(QF\text{-}ER) \quad \frac{F_{qf} \rightarrow s =_\xi t}{F_{qf} \rightarrow r[s/x^\xi] =_\tau r[t/x^\xi]}$$

where F_{qf} is a quantifier-free *formula*, s, t are terms of type ξ , r is a term of type τ and $r[s/x^\xi]$ denotes the simultaneous substitution of s for all occurrences of x in r (and similarly with t).

Remark 2.15. As F_{qf} in the formulation can actually be a formula with free variables, the rule (QF-ER) actually allows one to derive the seemingly stronger rule

$$(\Sigma_1\text{-}ER) \quad \frac{\exists y^\sigma F_{qf}(y) \rightarrow s =_\xi t}{\exists y^\sigma F_{qf}(y) \rightarrow r[s/x^\xi] =_\tau r[t/x^\xi]}$$

with F_{qf}, s, t, ξ, τ as before and where σ is an additional finite type and y is a variable of that type that is not free in r, s, t (see also [76]).

As is well-known, the combinators can be used to define λ -abstractions in the sense that for any term t of type τ and any variable x^ξ of type ξ , there is a term $\lambda x^\xi. t$ of type $\tau(\xi)$ such that provably

$$(\lambda x^\xi. t)(s^\xi) =_\tau t[s/x].$$

The system $\mathcal{A}^\omega$ then extends WE-PA^ω by the quantifier-free version of the axiom of choice in finite types, i.e.

$$(QF\text{-}AC) \quad \forall \underline{x} \exists \underline{y} F_{qf}(\underline{x}, \underline{y}) \rightarrow \exists \underline{Y} \forall \underline{x} F_{qf}(\underline{x}, \underline{Y}\underline{x})$$

where F_{qf} is quantifier-free and the $\underline{x}, \underline{y}$ may be of arbitrary type, and the axiom of dependent choice $DC = \{DC^\xi \mid \xi \subseteq T\}$ where

$$(DC^\xi) \quad \forall x^0, y^\xi \exists \underline{z}^\xi F(x, y, \underline{z}) \rightarrow \exists \underline{f}^{\xi(0)} \forall x^0 F(x, \underline{f}(x), \underline{f}(S(x)))$$

and where F is now of arbitrary complexity.

In the language of $\text{WE-PA}^\omega/\mathcal{A}^\omega$, rational and real numbers are represented as usual by objects of type 0 and 1, respectively, where we follow the conventions of [40] and here only briefly discuss the facts crucial to the presentation of the present paper. Concretely, using such codes, the operations $+_{\mathbb{Q}}, \cdot_{\mathbb{Q}}, (\cdot)_{\mathbb{Q}}^{-1}$ and relations $=_{\mathbb{Q}}, <_{\mathbb{Q}}$ are easily definable, with the latter given by quantifier-free formulas, and bootstrapped on that, one can introduce Π_1^0/Σ_1^0 -formulas $=_{\mathbb{R}}/<_{\mathbb{R}}$ on type 1 objects, which define the corresponding relations of the real numbers represented by the inputs, as well as closed terms $+_{\mathbb{R}}, \cdot_{\mathbb{R}}, |\cdot|_{\mathbb{R}}$ representing the usual operations of real arithmetic on these type 1 objects. The reciprocal $(\cdot)^{-1}$ on the reals, which crucially features in the later theory of monotone operators, is slightly more delicate to deal with as

there in fact is no closed term of type $1(1)$ in WE-PA^ω which represents r^{-1} correctly for all $r \neq 0$. Following [38], we handle this by using a binary term $(\cdot)^{-1}$ of type $1(1)(0)$ such that $(r)_l^{-1}$ correctly represents r^{-1} whenever $|r| > 2^{-l}$. In the following, we omit the subscripts of the arithmetical operations for $\mathbb{R}$ to avoid notational overload.

In the context of the bound extraction theorems established later, we associate a canonical type 1 representation $(r)_\circ$ with any real $r \in \mathbb{R}$.⁴ We omit the precise details of this operation and refer to e.g. [77] for the definition thereof, which anyhow is an immediate extension of the analogous operation only defined on the non-negative reals as originally introduced in [39] (see also [26, 40]). For the purpose of this paper, we only note the following lemma containing exactly the properties that we later need for this notion to be useful in the context of majorizability (trivially extending Lemma 2.10 from [39] to the full reals):

Lemma 2.16 (see e.g. [77]). *Let $r \in \mathbb{R}$. Then:*

- (1) $(r)_\circ$ is a representation of r in the sense of the above (see again e.g. [40]).
- (2) For $s \in [0, \infty)$, if $|r| \leq s$, then $(r)_\circ \leq_1 (s)_\circ$ (with $\leq_1$ as defined in e.g. [40]).
- (3) $(r)_\circ$ is nondecreasing (as a type 1 functional).

Lastly, in the context of the metatheorems, we will write r_α for the real represented by some type 1 functional α similar as in [40].

2.2.2. A system for abstract normed spaces. In this section, we briefly discuss the extension $\mathcal{A}^\omega[X, \|\cdot\|]$ of $\mathcal{A}^\omega$ as introduced in [26, 39] which provides a suitable system for the treatment of abstract normed spaces, extended later to treat corresponding dual spaces and notions from convex analysis. All these approaches crucially utilize the seminal paradigm of abstract types introduced in [39] that is prevalent in all modern approaches to proof mining. Namely, while the first logical metatheorems in proof mining relied on bare systems of arithmetic in all finite types and consequently only covered applications involving representable Polish metric spaces, these additional abstract types allow for the treatment of spaces which are not separable and thus not representable in the (pure) language of finite type arithmetic.

As such, the system $\mathcal{A}^\omega[X, \|\cdot\|]$ operates over an extended set of types T^X defined by

$$0, X \in T^X, \quad \xi, \tau \in T^X \Rightarrow \tau(\xi) \in T^X$$

where the additional abstract type X can then be utilized to represent an abstract and generic (in general non-separable) normed space. For that, we add additional constants to the resulting extended language to induce a linear and normed structure on X . Concretely, we add the constants $0_X, 1_X$ of type X , $+_X$ of type $X(X)(X)$, $-_X$ of type $X(X)$, $\cdot_X$ of type $X(X)(1)$ and $\|\cdot\|_X$ of type $1(X)$. It should be noted that $=_0$ is still the only primitive relation and in particular, identity on X is treated as a defined predicate via

$$x^X =_X y^X := \|x -_X y\|_X =_{\mathbb{R}} 0,$$

with equality at higher types defined similar to before. The theory $\mathcal{A}^\omega[X, \|\cdot\|]$ then arises from $\mathcal{A}^\omega$ by formulating the latter over the resulting extended language by extending the constants (if appropriate) to take arguments and produce values in those new types and by trivially extending the axiom schemes and rules to allow formulas from the new language and then adding the above new constants related to X together with the relevant defining axioms stating that X with these operations is a real normed vector space with 1_X such that $\|1_X\|_X =_{\mathbb{R}} 1$ and $-_X x$ being the additive inverse of x (see [26, 39, 40] for further details on all of this). Note that the extensionality of all those operations is provable in $\mathcal{A}^\omega[X, \|\cdot\|]$.

⁴Such an association will be non-effective but it will suffice for all intents and purposes that it behaves nice enough w.r.t. majorization.

2.2.3. *Systems for dual spaces, uniformly Fréchet differentiable functions and conjugates.* The system $\mathcal{D}^\omega$ introduced in [77] now extends the system $\mathcal{A}^\omega[X, \|\cdot\|]$ with an additional abstract type X^* used to intensionally and abstractly specify the dual of the space represented by X . We refer to [77] for a discussion of the underlying intuition and here only briefly present the main technical aspects of the theory as they are relevant to the present paper. At first, the system $\mathcal{D}^\omega$ operates over a further extended set of types T^{X, X^*} defined by

$$0, X, X^* \in T^{X, X^*}, \quad \xi, \tau \in T^{X, X^*} \Rightarrow \tau(\xi) \in T^{X, X^*}$$

and the resulting extended finite-type language is further enriched by a selection of constants restoring the linear and normed structure of X^* , similar as with X . Concretely, using this new type, we add constants $+_{X^*}$ of type $X^*(X^*)(X^*)$, $-_{X^*}$ of type $X^*(X^*)$, $\cdot_{X^*}$ of type $X^*(X^*)(1)$, 0_{X^*} and 1_{X^*} of type X^* and $\|\cdot\|_{X^*}$ of type $1(X^*)$ as before and, crucially, another constant $\langle \cdot, \cdot \rangle_{X^*}$ of type $1(X)(X^*)$ which serves as an abstract application functional, restoring the application character of X^* relative to X (similar as in the theory of topological vector spaces, see [77] for a further discussion). We do not add further relations and so, still, equality is a defined notion at types other than 0, with $=_X$ defined as before, $=_{X^*}$ defined via

$$x^* =_{X^*} y^* := \|x^* -_{X^*} y^*\|_{X^*} =_{\mathbb{R}} 0$$

and the higher-type equality defined similar to before.

The system $\mathcal{D}^\omega$, and all systems considered in this paper for that matter, result from $\mathcal{A}^\omega[X, \|\cdot\|]$ as formulated over that extended language by adding suitable additional axioms and rules (and even further constants in the case of a few systems discussed later). In that context, it should be noted that all the axioms that are being considered, and similarly all conclusions of rules, are of a particular logical form that later guarantees that these axioms and rules are admissible in the context of proof-theoretic bound extraction theorems in the style of proof mining. Concretely, all axioms considered here are “at most” of type Δ as originally introduced in [35, 36] (and then lifted to abstract types in [28]): A formula of type Δ is any formula of the form

$$\forall \underline{a}^\delta \exists \underline{b} \leq_\sigma \underline{r} \forall \underline{c}^\gamma F_{qf}(\underline{a}, \underline{b}, \underline{c})$$

where F_{qf} is quantifier-free, the types in $\underline{\delta}$, $\underline{\sigma}$ and $\underline{\gamma}$ are “low enough” (later made formal by the notion of admissible types, see Section 5 for further details), which in particular includes X and X^* , $\underline{r}$ is a tuple of closed terms of appropriate type and where $\leq$ is defined by recursion on the type via

- (1) $x \leq_0 y := x \leq_0 y$,
- (2) $x \leq_X y := \|x\|_X \leq_{\mathbb{R}} \|y\|_X$,
- (3) $x^* \leq_{X^*} y^* := \|x^*\|_{X^*} \leq_{\mathbb{R}} \|y^*\|_{X^*}$,
- (4) $x \leq_{\tau(\xi)} y := \forall z^\xi (xz \leq_\tau yz)$,

and with $\underline{x} \leq_\sigma \underline{y}$ being an abbreviation for $x_1 \leq_{\sigma_1} y_1 \wedge \dots \wedge x_k \leq_{\sigma_k} y_k$ where $\underline{x}$, $\underline{y}$ and $\underline{\sigma}$ are k -tuples of terms and types, respectively, such that x_i , y_i are of type σ_i . While encompassing a multitude of highly nontrivial mathematical statements, as in particular also evidenced by the axioms discussed in this paper, these sentences nevertheless have a very simple monotone functional interpretation and hence are admissible as axioms as will also be discussed in further detail later on.

In any way, the first crucial group of axioms for the system $\mathcal{D}^\omega$ that we consider is the following pair

$$(*)_{1,2} \quad \begin{cases} \forall x^{X^*}, x^X (|\langle x, x^* \rangle_{X^*}| \leq_{\mathbb{R}} \|x^*\|_{X^*} \|x\|_X), \\ \forall x^{X^*}, k^0 \exists x \leq_X 1_X (\|x^*\|_{X^*} - 2^{-k} \leq_{\mathbb{R}} |\langle x, x^* \rangle_{X^*}|), \end{cases}$$

which specify the norm of X^* as coded by $\|\cdot\|_{X^*}$ to be the actual supremum of $\langle x, x^* \rangle_{X^*}$ over the unit ball.⁵

For a second crucial group of axioms, we further specify the application functional to be a bilinear map:

$$\begin{aligned}
 (*)_3 \quad & \begin{cases} \forall x^X, x^{*X^*}, y^{*X^*}, \alpha^1, \beta^1 (\langle x, \alpha x^* +_{X^*} \beta y^* \rangle_{X^*} =_{\mathbb{R}} \alpha \langle x, x^* \rangle_{X^*} + \beta \langle x, y^* \rangle_{X^*}), \\ \forall x^X, x^{*X^*}, y^{*X^*}, \alpha^1, \beta^1 (\langle x, \alpha x^* -_{X^*} \beta y^* \rangle_{X^*} =_{\mathbb{R}} \alpha \langle x, x^* \rangle_{X^*} - \beta \langle x, y^* \rangle_{X^*}), \end{cases} \\
 (*)_4 \quad & \begin{cases} \forall x^X, y^X, x^{*X^*}, \alpha^1, \beta^1 (\langle \alpha x +_X \beta y, x^* \rangle_{X^*} =_{\mathbb{R}} \alpha \langle x, x^* \rangle_{X^*} + \beta \langle y, x^* \rangle_{X^*}), \\ \forall x^X, y^X, x^{*X^*}, \alpha^1, \beta^1 (\langle \alpha x -_X \beta y, x^* \rangle_{X^*} =_{\mathbb{R}} \alpha \langle x, x^* \rangle_{X^*} - \beta \langle y, x^* \rangle_{X^*}). \end{cases}
 \end{aligned}$$

Also, as a third group of axioms $(*)_5$, we add the vector space axioms for X^* formulated with the operations $+_{X^*}$, $-_{X^*}$, $\cdot_{X^*}$, 0_{X^*} , 1_{X^*} w.r.t. $=_{X^*}$ and we also axiomatically specify $\|1_{X^*}\|_{X^*} =_{\mathbb{R}} 1$.

The last additions to the system concern the relationship between the bounded linear functionals represented in $1(X)$ (where elements of the dual of X , if naively specified, live) and the elements of X^* . While it will of course not be permissible meanwhile aiming for bound extraction theorems to include an axiom that guarantees the existence of a representing element in X^* for every element of $1(X)$ which is a continuous linear functional (which is why the dual X^* was intensionally approached in the first place, see [77] for a further discussion), we resort to the next best thing available in this situation by including a rule guaranteeing that at least all terms of type $1(X)$ which provably are continuous linear functionals indeed are represented by a corresponding element of type X^* and thus consider the rule

$$(\text{QF-LR}) \quad \frac{F_{qf} \rightarrow (\forall x^X, y^X, \alpha^1, \beta^1 (t(\alpha x +_X \beta y) =_{\mathbb{R}} \alpha tx + \beta ty) \wedge \forall x^X (|tx| \leq_{\mathbb{R}} M \|x\|_X))}{F_{qf} \rightarrow \exists x^* \leq_{X^*} M 1_{X^*} \forall x^X (tx =_{\mathbb{R}} \langle x, x^* \rangle_{X^*})}$$

where F_{qf} is a quantifier-free formula and where t and M are terms of type $1(X)$ and 1 , respectively. Also, to ease formal development of the theory of X^* , we further axiomatically populate X^* by adding the crucial consequence of the Hahn-Banach theorem that the normalized duality map J discussed previously has non-empty values, i.e. that $J(x) \neq \emptyset$ for $x \in X$, which we achieve by the following axiom:

$$(*)_6 \quad \forall x^X \exists x^* \leq_{X^*} \|x\|_X 1_{X^*} (\langle x, x^* \rangle_{X^*} =_{\mathbb{R}} \|x\|_X^2 =_{\mathbb{R}} \|x^*\|_{X^*}^2).$$

The system $\mathcal{D}^\omega$ for the abstract dual space of an abstract normed space is now officially defined as the extension of $\mathcal{A}^\omega[X, \|\cdot\|]$, formulated over the extended language using the types T^{X, X^*} , by the above constants, axioms and rules.

Various essential properties of the space X^* are immediately provable in the system (including simple but crucial aspects like that $\|\cdot\|_{X^*}$ is indeed a norm) and we refer to [77] for further details.

Over this system, convex functions are then introduced as objects of type $1(X)$ and we in particular consider a system where a constant f of type $1(X)$ is added to the language which is axiomatically specified to be convex via the axiom⁶

$$(f)_1 \quad \forall x^X, y^X, \lambda^1 \left(f \left(\tilde{\lambda} x +_X (1 - \tilde{\lambda}) y \right) \leq_{\mathbb{R}} \tilde{\lambda} f(x) + (1 - \tilde{\lambda}) f(y) \right).$$

In that way, for the system which we are in the process of specifying, and throughout this paper for that matter, we will only consider convex functions that are total, i.e. where $\text{dom } f = X$.

⁵These axioms actually arise through a general approach to treating suprema over certain bounded sets as introduced in [77] and we refer to that paper for a further discussion on the particularities and practicalities of these axioms.

⁶Here, $\tilde{\cdot}$ is defined as e.g. in [40], allowing for implicit quantification over $[0, 1]$.

We refer to [77] for some discussions on how non-total convex functions might be approached formally.

Gradients of uniformly Fréchet differentiable functions are then introduced via introducing suitably continuous selection functions of the subgradient, utilizing the characterization results presented in Propositions 2.3 and 2.4. Concretely, we in that way treat a uniformly Fréchet differentiable f with a (total) gradient by adding another constant ∇f of type $X^*(X)$ to the system together with the axioms that ∇f is a selection of ∂f , i.e.

$$(\nabla f)_1 \quad \forall x^X, y^X (f(x) + \langle y -_X x, \nabla f(x) \rangle_{X^*} \leq_{\mathbb{R}} f(y)),$$

and that ∇f is uniformly continuous on bounded subsets, i.e.

$$(\nabla f)_2 \quad \forall x^X, y^X, b^0, k^0 \left(\|x\|_X, \|y\|_X <_{\mathbb{R}} b \wedge \|x -_X y\|_X <_{\mathbb{R}} 2^{-\omega^{\nabla f}(k,b)} \right. \\ \left. \rightarrow \|\nabla f(x) -_{X^*} \nabla f(y)\|_{X^*} \leq_{\mathbb{R}} 2^{-k} \right),$$

where $\omega^{\nabla f}$ is another additional constant of type $0(0)(0)$ representing the modulus of uniform continuity.

We denote the theory resulting from $\mathcal{D}^\omega$ by extending it with the previous constants and axioms by $\mathcal{D}^\omega[f, \nabla f]$.

Crucially, we now also want to treat the associated Fenchel conjugate f^* and if this map is to be treated in any way amenable to the metatheorems, it has to be majorizable which means that it in particular has to be bounded on bounded sets. By the result collected in Proposition 2.5, this is exactly the case if, and only if, f is supercoercive in which case f^* is total. In that case however, as shown in [77], if f is supercoercive, then the supremum defining it is already attained over a suitably large ball around on the origin and that corresponding radius can be computed in terms of a given modulus witnessing the supercoercivity of f quantitatively:

Lemma 2.17 (essentially [77]). *Let $\alpha : \mathbb{N} \rightarrow \mathbb{N}$ be a modulus of supercoercivity, i.e.*

$$\forall K \in \mathbb{N}, x \in X (\|x\| > \alpha(K) \rightarrow f(x) \geq K \|x\|)$$

and let $c \geq |f(0)|$. Then for $x^ \in X^*$ with $\|x^*\| \leq b$, we have*

$$f^*(x^*) = \sup_{x \in \overline{B}_{r(\alpha, c, b)}(0)} (\langle x, x^* \rangle - f(x))$$

where $r(\alpha, c, b) = \max\{\alpha(b+1) + 1, c+1\}$.

This in turn allows one to utilize the abstract approach developed in [77] on the formal treatment of suprema over certain bounded sets in systems amenable for proof mining to introduce corresponding axioms that, given another constant f^* of type $1(X^*)$, specify that this constant is really the conjugate corresponding to f (we again refer to [77] for a further discussion of this): The first axiom specifies that f is supercoercive with modulus α^f via

$$(f)_2 \quad \forall K^0, x^X (\|x\|_X >_{\mathbb{R}} \alpha^f(K) \rightarrow f(x) \geq_{\mathbb{R}} K \|x\|_X)$$

where α^f is an additional constant of type 1. The two further axioms specify that f^* is actually given by the respective supremum which is achieved by first specifying that f^* is a pointwise upper bound for all affine functionals $g_x(x^*) = \langle x, x^* \rangle - f(x)$ via

$$(f^*)_1 \quad \forall x^{*X^*}, x^X (\langle x, x^* \rangle_{X^*} - f(x) \leq_{\mathbb{R}} f^*(x^*))$$

and another axioms specifying that f^* is indeed the pointwise supremum of all these affine functionals via

$$(f^*)_2 \quad \forall x^{*X^*}, b^0, k^0 \exists x \leq_X \max\{\alpha^f(b+1) + 1, [|f(0)|](0) + 2\} 1_X \\ (\|x^*\|_{X^*} <_{\mathbb{R}} b \rightarrow (f^*(x^*) - 2^{-k} \leq_{\mathbb{R}} \langle x, x^* \rangle_{X^*} - f(x))).$$

Here, $\max\{\alpha^f(b+1) + 1, [|f(0)|](0) + 2\}$ immediately arises as a bound restricting the defining supremum of f^* via Lemma 2.17 since $[|f(0)|](0) + 1 \geq |f(0)|$.

If f^* is uniformly Fréchet differentiable on bounded sets as well, its gradient can now be introduced as before by adding a constant ∇f^* of type $X(X^*)$ together with the axioms that ∇f^* is a selection of ∂f^* , i.e.

$$(\nabla f^*)_1 \quad \forall x^{*X^*}, y^{*X^*} (f^*(x^*) + \langle \nabla f^*(x^*), y^* -_{X^*} x^* \rangle_{X^*} \leq_{\mathbb{R}} f^*(y^*)),$$

and that ∇f^* is uniformly continuous on bounded subsets, i.e.

$$(\nabla f^*)_2 \quad \forall x^{*X^*}, y^{*X^*}, b^0, k^0 \left(\|x^*\|_{X^*}, \|y^*\|_{X^*} <_{\mathbb{R}} b \wedge \|x^* -_{X^*} y^*\|_{X^*} <_{\mathbb{R}} 2^{-\omega^{\nabla f^*}(k,b)} \right. \\ \left. \rightarrow \|\nabla f^*(x^*) -_X \nabla f^*(y^*)\|_X \leq_{\mathbb{R}} 2^{-k} \right),$$

where $\omega^{\nabla f^*}$ is another additional constant of type $0(0)(0)$, coding the respective modulus of uniform continuity.

In the context of the above axioms, whereby the functions f and f^* are Fréchet differentiable, Proposition 2.7 yields that the function f is a Legendre function. Here note that, by the results of [11, 12], any space X where such a convex function f exists is immediately (super-)reflexive, so that Proposition 2.7 applies. Therefore, the gradients ∇f and ∇f^* are inverses of each other. Correspondingly, we add this fact as a last axiom

$$(L) \quad \forall x^X, x^{*X^*} (\nabla f(\nabla f^*(x^*)) =_{X^*} x^* \wedge \nabla f^*(\nabla f(x)) =_X x).$$

We write $\mathcal{D}^\omega[f, \nabla f, f^*, \nabla f^*]$ for the system $\mathcal{D}^\omega[f, \nabla f]$ extended with the above constants and axioms regarding the function f^* and its gradient ∇f^* .

We finally just quote a few properties that are immediately provable in this system:

Lemma 2.18 ([77]). *The theory $\mathcal{D}^\omega[f, \nabla f]$ proves:*

(1) *f is uniformly Fréchet differentiable on bounded subsets, i.e.*

$$\forall b^0, k^0 \exists j^0 \forall x^X, y^X (\|x\|_X <_{\mathbb{R}} b \wedge 0 <_{\mathbb{R}} \|y\|_X <_{\mathbb{R}} 2^{-j} \\ \rightarrow |f(x+y) - f(x) - \langle y, \nabla f(x) \rangle_{X^*}| \leq_{\mathbb{R}} 2^{-k} \|y\|_X),$$

where in fact one can choose $j = \omega^{\nabla f}(k, b+1)$.

(2) *∇f is bounded on bounded subsets, i.e.*

$$\forall b^0 \exists c^0 \forall x^X (\|x\|_X <_{\mathbb{R}} b \rightarrow \|\nabla f(x)\|_{X^*} \leq_{\mathbb{R}} c),$$

where in fact one can choose $c = C(b) = b2^{\omega^{\nabla f}(0,b)} + [|\nabla f(0)|]_{X^*}(0) + 2$.

(3) *f is uniformly continuous on bounded subsets, i.e.*

$$\forall k^0, b^0 \exists j^0 \forall x^X, y^X (\|x\|_X, \|y\|_X <_{\mathbb{R}} b \wedge \|x -_X y\|_X \leq_{\mathbb{R}} 2^{-j} \\ \rightarrow |f(x) - f(y)| \leq_{\mathbb{R}} 2^{-k}),$$

where in fact one can choose $j = \omega^f(k, b) = k + C(b)$.

(4) f is bounded on bounded sets, i.e.

$$\forall b^0 \exists d^0 \forall x^X (\|x\|_X <_{\mathbb{R}} b \rightarrow |f(x)| \leq_{\mathbb{R}} d),$$

where in fact one can choose $d = D(b) = b2^{\omega^f(0,b)} + [|f(0)|](0) + 2$.

Correspondingly, the theory $\mathcal{D}^\omega[f, \nabla f, f^*, \nabla f^*]$ proves similar properties appropriately formulated for f^* and ∇f^* . Also, this latter system proves that f^* is convex.

Also, clearly, the Bregman distance

$$D_f(x, y) := f(x) - f(y) - \langle x - y, \nabla f(y) \rangle$$

and its associated dual

$$W_f(x, x^*) := f(x) - \langle x, x^* \rangle + f^*(x^*)$$

discussed before can now just be given via closed terms in the language of $\mathcal{D}^\omega[f, \nabla f, f^*, \nabla f^*]$ and this system immediately proves the so-called three and four point identities (see e.g. [6]):

Lemma 2.19 ([77]). *The system $\mathcal{D}^\omega[f, \nabla f, f^*, \nabla f^*]$ proves the three and four point identities:*

$$\begin{aligned} (1) \quad & \left\{ \begin{aligned} & \forall x^X, y^X, z^X (D_f(x, y) + D_f(y, z) - D_f(x, z) \\ & \quad =_{\mathbb{R}} \langle x -_X y, \nabla f(z) -_{X^*} \nabla f(y) \rangle_{X^*}). \end{aligned} \right. \\ (2) \quad & \left\{ \begin{aligned} & \forall x^X, y^X, z^X, w^X (D_f(y, x) - D_f(y, z) - D_f(w, x) + D_f(w, z) \\ & \quad =_{\mathbb{R}} \langle y -_X w, \nabla f(z) -_{X^*} \nabla f(x) \rangle_{X^*}). \end{aligned} \right. \end{aligned}$$

The system $\mathcal{D}^\omega[f, \nabla f, f^*, \nabla f^*]$ requires various properties of the associated functions and, corresponding to that, features various key moduli. The following example discusses how these moduli can be instantiated for some of the particular cases surveyed in Example 2.8:

Example 2.20. We focus on the rather canonical but nevertheless key and illustrative case (recall Example 2.8) of $f(x) := \frac{1}{p} \|x\|^p$ for $p \in (1, \infty)$ over a uniformly convex and uniformly smooth (and hence reflexive) Banach space $(X, \|\cdot\|)$. By Lemma 6.2 in [5], it follows that $f^*(x^*) = \frac{1}{q} \|x^*\|^q$ for q such that $\frac{1}{p} + \frac{1}{q} = 1$ as well as that f and f^* are Fréchet differentiable with $\nabla f(x) = \|x\|^{p-2} J(x)$ as well as $\nabla f^*(x^*) = \|x^*\|^{q-2} J^*(x^*)$, where we wrote J and $J^* = J^{-1}$ for the (single-valued) normalized duality maps of X and X^* , respectively, defined as before.

As $p > 1$, we get that f is supercoercive with modulus $\alpha^f(K) = \sqrt[p-1]{pK}$. Moduli of uniform continuity on bounded sets for ∇f and ∇f^* can be constructed from associated moduli of uniform convexity for the spaces X and X^* (recall that X^* is uniformly convex if, and only if, X is uniformly smooth). Concretely, let $\eta : \mathbb{N} \rightarrow \mathbb{N}$ be a modulus of uniform convexity for X , i.e.

$$\|x - y\| \geq 2^{-k} \text{ implies } \left\| \frac{1}{2}(x + y) \right\| \leq 1 - 2^{-\eta(k)}$$

for all $x, y \in X$ with $\|x\|, \|y\| \leq 1$ and all $k \in \mathbb{N}$, and let $\eta^* : \mathbb{N} \rightarrow \mathbb{N}$ be a similar such modulus for X^* . These moduli can be easily instantiated if X is a Hilbert space, where then $\eta(k) := 2k + 3$ (which e.g. follows from Proposition 8 in [61]). Other cases include in particular L^r -spaces with $r \in (1, \infty)$, where corresponding formulas can be found in [65] (see also [49]). Since X^* is uniformly convex with a modulus η^* , it follows from Proposition 2.5 of [49] (which in turn relies on well-known results due to Lindenstrauss and Tzafriri [65]) that J is uniformly continuous on bounded sets with modulus $\omega(k, b) = k + 2 + \eta^*(k + \max\{b, 1\})$. By the same result, as $X = X^{**}$ is uniformly convex with modulus η , we get that J^* is uniformly continuous on bounded sets with modulus $\omega^*(k, b) = k + 2 + \eta(k + \max\{b, 1\})$.

From these moduli for J and J^* , we can rather immediately derive corresponding moduli of uniform continuity on bounded sets for ∇f and ∇f^* , respectively. First note that J is bounded

on bounded sets, with the estimate $\|J(x)\| \leq b2^{\omega(0,b)}$ for $x \in X$ with $\|x\| \leq b$ (which can be derived similar to e.g. item (2) of Lemma 2.18). Hence, a rather direct calculation yields

$$\begin{aligned} \|\|x\|^{p-2} J(x) - \|y\|^{p-2} J(y)\| &\leq |\|x\|^{p-2} - \|y\|^{p-2}| \|J(x)\| + \|y\|^{p-2} \|J(x) - J(y)\| \\ &\leq |\|x\|^{p-2} - \|y\|^{p-2}| \cdot b2^{\omega(0,b)} + b^{p-2} \|J(x) - J(y)\| \end{aligned}$$

for any $x, y \in X$ with $\|x\|, \|y\| \leq b$. Now, if $p > 2$, then $|\|x\|^{p-2} - \|y\|^{p-2}| \leq (p-2)b^{p-3} \|x - y\|$ and if $p \in (1, 2]$, then $|\|x\|^{p-2} - \|y\|^{p-2}| \leq \|x - y\|^{p-2}$. Combined, we get that

$$\omega^{\nabla f}(k, b) := \max\{\omega(k+1 + \lceil b^{p-2} \rceil, b), k+1 + \lceil (p-2)b^{p-2} \rceil \omega(0, b)\}$$

is a modulus of uniform continuity on bounded sets for ∇f whenever $p > 2$, and

$$\omega^{\nabla f}(k, b) := \max\{\omega(k+1 + \lceil b^{p-2} \rceil, b), \lceil (k+1 + b + \omega(0, b))/(p-2) \rceil\}$$

is a modulus of uniform continuity on bounded sets for ∇f whenever $p \in (1, 2]$. Exchanging ω with ω^* and p with q yields a modulus of uniform continuity on bounded sets $\omega^{\nabla f^*}$ for ∇f^* .

3. LOGICAL SYSTEMS FOR OPERATORS AND THEIR RESOLVENTS

We now turn to the main aim of this paper, which is to extend the previously discussed systems for convex functions, their derivatives and conjugates over Banach spaces with their duals with a treatment for monotone operators on Banach spaces in the sense of Browder as introduced before, together with the relativized resolvents in the style of Eckstein, both of which are approach similarly as monotone operators and their usual resolvents in Hilbert spaces as developed in [76].

The corresponding basic system $\mathcal{D}^\omega[f, \nabla f, f^*, \nabla f^*]$ and its axioms require various properties of the associated function, and as such we from now on make the following standing assumption regarding any function f considered:

Assumption 3.1. In the following, we work over a Banach space $(X, \|\cdot\|)$ together with a convex and supercoercive function $f : X \rightarrow \mathbb{R}$ with Fenchel conjugate $f^* : X^* \rightarrow \mathbb{R}$ such that both f and f^* are Fréchet differentiable everywhere with a gradients ∇f and ∇f^* which are uniformly continuous on bounded sets.

Quantitatively, we require access to moduli of uniform continuity on bounded sets $\omega^{\nabla f}, \omega^{\nabla f^*} : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ for ∇f and ∇f^* , respectively, as well as a modulus of supercoercivity $\alpha^f : \mathbb{N} \rightarrow \mathbb{N}$ for f .

Recall that, as already mentioned previously, by the results of [11, 12] any space X where a convex function f exists that satisfies the above Assumption 3.1 is immediately (super-)reflexive.

The moduli $\omega^{\nabla f}$ and $\omega^{\nabla f^*}$ as well as α^f fixed above are key numerical data in the later metatheorems. In particular, beyond the fact that they allow for an inclusion of the respective properties as axioms in the formal system in a tame way, they also ensure the majorizability of ∇f and f as well as ∇f^* and f^* (as already in [77]) as well as related objects, in particular the resolvents of a monotone operator as treated in Section 3.2 later on (see Lemma 5.6 later on). We refer to Example 2.20 above for how these moduli can be instantiated in some practical cases.

While we throughout work with (extensions of) the above system, and hence under this standing assumption, we want to highlight that, presumably, many of the considerations made here could be extended mutatis mutandis to the case where we only consider partial convex functions f with an intensional treatment of the domain as discussed in [77], but we do not discuss this any further.

Also, while the treatment for monotone operators over Hilbert spaces presented in [76] was divided on whether the resolvents are partial or total, we in the following will only consider systems for monotone operators on Banach spaces where the (relativized) resolvents are all total. In particular, as we working in the context of Assumption 3.1, Proposition 2.14 thereby yields that the totality of the resolvent is equivalent to the maximal monotonicity of the operator. Further, we will for convenience assume that the operator has a zero, which will prove very convenient for majorization purposes later on (see in particular the discussion in Section 3.2). We collect these requirements into another standing assumption:

Assumption 3.2. In the following, in addition to Assumption 3.1, we work with a maximally monotone operator $A : X \rightarrow 2^{X^*}$, with corresponding f -resolvents $\text{Res}_{\gamma A}^f$ for $\gamma > 0$, such that $A^{-1}0 \neq \emptyset$.

This treatment of maximally monotone operators via their total (relativized) resolvents, that is via Proposition 2.14, is hence analogous to the approach to maximally monotone operators in Hilbert spaces taken in [76], and in particular leads to a proof-theoretically tame system for such objects while direct considerations of maximality are in general not feasible (as will be explored further in Section 4 later on). Also, we want to remark that if one would want to treat operators with partial resolvents, then a similar approach as presented in [76] could presumably also be followed here.

In the following, we will omit types of variables whenever convenient and omit types in proofs almost always to make everything more readable.

3.1. Further considerations on convex functions. As mentioned before, the basic system for all extensions considered here will be $\mathcal{D}^\omega[f, \nabla f, f^*, \nabla f^*]$ from the preceding section, treating the dual of the abstract normed space together with a convex function, its Fenchel-conjugate and their uniformly continuous gradients.

It will however be convenient to slightly extend this system so that the theory of monotone operators can be developed smoothly. Concretely, it will be convenient to axiomatically include a few more properties of a convex function and its Fenchel conjugate into the previous system. Namely, by the Fenchel–Moreau theorem (see e.g. [10]), we know that if f is proper, lower-semicontinuous and convex, then f^* is proper and $f = f^{**}$ where we define $f^{**} : X \rightarrow (-\infty, +\infty]$ by

$$f^{**}(x) := \sup_{x^* \in X^*} (\langle x, x^* \rangle - f^*(x^*)).$$

With this definition, we follow one particular approach to biconjugates as e.g. outlined in [10]. In different works, one also finds f^{**} introduced as $(f^*)^*$ acting on X^{**} and thus on X by its embedding into X^{**} , which anyhow coincides with X in the context of reflexivity, so that these different approaches yield the same object in our context, but the above formulation will also influence the types of objects considered later.

Naturally, a function f as axiomatized by the system $\mathcal{D}^\omega[f, \nabla f, f^*, \nabla f^*]$ satisfies the assumptions of the Fenchel–Moreau theorem (recall Assumption 3.1) and the resulting fact that $f = f^{**}$ is crucial for the development of the theory of monotone operators, so we need to deal with it formally. However, we do not treat this fact by analyzing the corresponding proof and verifying that it indeed can be carried out in $\mathcal{D}^\omega[f, \nabla f, f^*, \nabla f^*]$ but instead hardwire it into the system akin to how f^* is treated in $\mathcal{D}^\omega[f, \nabla f, f^*, \nabla f^*]$.

In more detail, note that, in the context of the above systems, $f = f^{**} = (f^*)^*$ is bounded on bounded sets and therefore f^* is supercoercive by Proposition 2.5. In fact, by analyzing the proof of Proposition 2.5 as e.g. given in [5], we find that a corresponding modulus of supercoercivity for f^* can be computed from a function witnessing that f is bounded on bounded

sets which we briefly sketch in Remark 3.3 below. In any way, in the presence of a modulus of supercoercivity α^{f^*} for f^* , the fact that $f = f^{**}$ can be wired into the system with the following axioms which are analogous to the axioms $(f^*)_1, (f^*)_2$ from the preceding section (and which similarly instantiate the axiom schemes introduced in [77] for treating suprema over certain bounded sets) where the corresponding bound in the axiom $(f^{**})_2$ again is derived via Lemma 2.17:

$(f^*)_3$ f^* is supercoercive with modulus α^{f^*} , i.e.

$$\forall K^0, x^{*X^*} (\|x^*\|_{X^*} >_{\mathbb{R}} \alpha^{f^*}(K) \rightarrow f^*(x^*) \geq_{\mathbb{R}} K \|x^*\|_{X^*}).$$

Here, α^{f^*} is an additional constant of type 1.

$(f^{**})_1$ f is the pointwise upper bound for all affine functionals $g_{x^*}(x) = \langle x, x^* \rangle - f^*(x^*)$, i.e.

$$\forall x^X, x^{*X^*} (\langle x, x^* \rangle_{X^*} - f^*(x^*) \leq_{\mathbb{R}} f(x)).$$

$(f^{**})_2$ f is indeed the pointwise supremum of these affine functionals, i.e.

$$\begin{aligned} \forall x^X, b^0, k^0 \exists x^* \leq_{X^*} \max\{\alpha^{f^*}(b+1) + 1, [|f^*(0)|](0) + 2\} 1_{X^*} \\ (\|x\|_X <_{\mathbb{R}} b \rightarrow (f(x) - 2^{-k} \leq_{\mathbb{R}} \langle x, x^* \rangle_{X^*} - f^*(x^*))) \end{aligned}$$

With $\mathcal{D}_{f, f^*}^\omega[\text{FM}]$ we abbreviate the system that arises from $\mathcal{D}^\omega[f, \nabla f, f^*, \nabla f^*]$ by adding these constants and axioms. As all these (new) axioms are again of type Δ as discussed before, they are also admissible while aiming for bound extraction theorems in the style of proof mining.

In analogy to α^f , also α^{f^*} later features as a key numerical datum in the context of the metatheorems (see again Lemma 5.6 later on). However, while we include such a modulus abstractly for convenience, as mentioned before it is essentially redundant as it can be explicitly computed from a modulus D for f being bounded on bounded sets, which in turn can be derived from $\omega^{\nabla f}$ (recall Lemma 2.18). This is illustrated in the following remark:

Remark 3.3. As shown in the proof of [5, Lemma 3.2] (which is the central ingredient of the proof of Proposition 2.5), given $\alpha > 0$ and $\beta \in \mathbb{R}$, one has $\alpha \|x\| + \beta \leq f(x)$ for any $x \in X$ if, and only if, $f^*(x^*) \leq \iota_{\overline{B}_\alpha(0)}(x^*) - \beta$ for any $x^* \in X^*$, with $\overline{B}_\alpha(0)$ seen in X^* and where ι_C is the indicator function of a set C defined by

$$\iota_C(z) := \begin{cases} 0 & \text{if } z \in C, \\ +\infty & \text{if } z \notin C. \end{cases}$$

If now $D : \mathbb{N} \rightarrow \mathbb{N}$ is a modulus for $f = f^{**}$ being bounded on bounded sets (as e.g. constructed in Lemma 2.18), then we immediately have $f^{**}(x) = f(x) \leq \iota_{\overline{B}_b(0)}(x) + D(b+1)$ for any non-zero $b \in \mathbb{N}$ and $x \in X$ so that the above equivalence yields $b \|x^*\| - D(b+1) \leq f^*(x^*)$ for any $x^* \in X^*$. In particular, for any $K \in \mathbb{N}$ and $x^* \neq 0$, we have

$$\|x^*\| \left(K + 1 - \frac{D(K+2)}{\|x^*\|} \right) \leq f^*(x^*)$$

so that, for $\|x^*\| > D(K+2)$, we have $K \|x^*\| \leq f^*(x^*)$. This yields that $\alpha(K) := D(K+2)$ is a modulus of supercoercivity for f^* as in Lemma 2.17.

One crucial aspect of the theory of the conjugate function f^* is that it provides alternative characterizations of continuity properties of ∇f in terms of convexity properties of f^* . These play a central role in the context of the theory of monotone operators in Banach spaces and so, before moving on to the latter, we first formally establish some of these corresponding convexity properties of f^* . As a byproduct, we also formally provide further quantitative properties of the gradient ∇f .

Lemma 3.4. *The system $\mathcal{D}^\omega[f, \nabla f, f^*, \nabla f^*]$ proves:*

(1) *The “Fenchel-Young equality”⁷ for any subgradient u^* of f at x , i.e.*

$$\forall x^X, u^{*X^*} (\forall y^X (f(y) \geq_{\mathbb{R}} f(x) + \langle y -_X x, u^* \rangle_{X^*}) \rightarrow f(x) + f^*(u^*) =_{\mathbb{R}} \langle x, u^* \rangle_{X^*}).$$

(2) *The “Fenchel-Young equality” for $\nabla f x$, i.e.*

$$\forall x^X (f(x) + f^*(\nabla f x) =_{\mathbb{R}} \langle x, \nabla f x \rangle_{X^*}).$$

(3) *Approximate subgradients of f are close to the gradient of f , i.e.*

$$\begin{aligned} \forall b^0, k^0 \exists j^0 \forall x^X, x^{*X^*} (\|x\|_X <_{\mathbb{R}} b \wedge \forall y^X (\langle y -_X x, x^* \rangle_{X^*} + f(x) \leq_{\mathbb{R}} f(y) + 2^{-j}) \\ \rightarrow \|x^* -_{X^*} \nabla f x\|_{X^*} \leq_{\mathbb{R}} 2^{-k}), \end{aligned}$$

where in fact we can take

$$j = k + 4 + \omega^{\nabla f}(k + 3, b + 1).$$

(4) *The “Fenchel-Young equality” characterizes gradients of f , i.e.*

$$\forall x^X, x^{*X^*} (f(x) + f^*(x^*) =_{\mathbb{R}} \langle x, x^* \rangle_{X^*} \rightarrow x^* =_{X^*} \nabla f x),$$

where in fact it moreover holds that

$$\begin{aligned} \forall b^0, k^0 \exists j^0 \forall x^X, x^{*X^*} (\|x\|_X <_{\mathbb{R}} b \wedge f(x) + f^*(x^*) - \langle x, x^* \rangle_{X^*} \leq_{\mathbb{R}} 2^{-j} \\ \rightarrow \|x^* -_{X^*} \nabla f x\|_{X^*} \leq_{\mathbb{R}} 2^{-k}) \end{aligned}$$

where we can take

$$j = k + 4 + \omega^{\nabla f}(k + 3, b + 1).$$

(5) *$\nabla f x$ is the unique subgradient of f at x , i.e.*

$$\forall x^X, u^{*X^*} (\forall y^X (f(y) \geq_{\mathbb{R}} f(x) + \langle y -_X x, u^* \rangle_{X^*}) \rightarrow u^* =_{X^*} \nabla f x).$$

(6) *f^* is uniformly strictly convex on bounded subsets, i.e.*

$$\begin{aligned} \forall k^0, i^0, b^0 \exists j^0 \forall x^{*X^*}, y^{*X^*}, t^1 (\|x^*\|_{X^*}, \|y^*\|_{X^*} <_{\mathbb{R}} b \wedge 2^{-i} \leq_{\mathbb{R}} t \leq_{\mathbb{R}} 1 - 2^{-i} \\ \wedge t f^*(x^*) + (1 - t) f^*(y^*) - f^*(tx^* +_{X^*} (1 - t)y^*) \leq_{\mathbb{R}} 2^{-j} \\ \rightarrow \|x^* -_{X^*} y^*\|_{X^*} \leq_{\mathbb{R}} 2^{-k}) \end{aligned}$$

where we in fact can choose

$$j = (k + 4 + \omega^{\nabla f}(k + 4, F(b) + 2)) + i$$

where F is a modulus for ∇f^* being bounded on bounded sets (which can be constructed similar to Lemma 2.18).

(7) *∇f^* is uniformly strictly monotone on bounded subsets, i.e.*

$$\begin{aligned} \forall k^0, b^0 \exists j^0 \forall x^{*X^*}, y^{*X^*} (\|x^*\|_{X^*}, \|y^*\|_{X^*} <_{\mathbb{R}} b \wedge \|x^* -_{X^*} y^*\|_{X^*} >_{\mathbb{R}} 2^{-k} \\ \rightarrow \langle \nabla f^* x^* -_X \nabla f^* y^*, x^* -_{X^*} y^* \rangle_{X^*} \geq_{\mathbb{R}} 2^{-j}). \end{aligned}$$

where we in fact can choose

$$j = k + 5 + \omega^{\nabla f}(k + 4, F(b) + 2).$$

with all other constants as in (6).

⁷By this expression, we mean in the following that the Fenchel-Young inequality is not strict, i.e. is satisfied with equality.

The proofs are essentially all careful implementations of standard arguments from the literature. The key items are (1), (3), (6) and (7), with (2), (4) and (5) being natural corollaries thereof. Item (1) is a simple calculation, and follows the same reasoning as the corresponding result in Hilbert spaces (see e.g. Proposition 16.10 in [7]). Item (3) presents a quantitative variant of the result that for a Fréchet differentiable function f , the gradient ∇f is the unique subgradient (recall Proposition 2.3). Indeed, the proof of item (3) arises as a combination of quantitative analyses of the usual uniqueness proofs as found in Hilbert spaces (see Proposition 17.31 in [7]) as well as Banach spaces (see Theorem 2.4.4 in [99]).

Item (6) requires the mathematically most involved argument, and implements one part of the equivalence between differentiability of a function and strict convexity of its conjugate as already mentioned previously (see e.g. Corollary 18.12 in [7] for such results already in the context of Hilbert spaces). Concretely, in our context, as the function f is Fréchet differentiable with a uniformly continuous gradient, it is also uniformly Frechet differentiable (recall Lemma 2.18), and so the conjugate f^* is uniformly strictly convex in the above sense, a notion going back to the work of Asplund [2] which is intimately related (see in particular Theorem 2.10 in [18]) to total convexity [16]. Our formal proof given here arises as a combination of arguments used in Hilbert spaces (see Proposition 17.31 in [7]) as well as Banach spaces (see [17] and [98]).

Item (7) is then derived from (6) by a variant of the canonical result that gradients of (uniformly) strictly convex functions are (uniformly) strictly monotone, which follows the same reasoning as usually made in Hilbert spaces (see e.g. Example 22.4 in [7]), suitably lifted to Banach spaces (see also Lemma 3.4 in [84]).

Proof of Lemma 3.4. (1) Let u^* be such that $\forall y (f(y) \geq f(x) + \langle y - x, u^* \rangle)$, i.e.

$$\langle y, u^* \rangle - f(y) \leq \langle x, u^* \rangle - f(x)$$

for all y . Using $(f^*)_2$, we get that for any j , there exists a y_j such that

$$f^*(u^*) - (\langle x, u^* \rangle - f(x)) \leq f^*(u^*) - (\langle y_j, u^* \rangle - f(y_j)) \leq 2^{-j}$$

and thus we have $f^*(u^*) \leq \langle x, u^* \rangle - f(x)$. Using axiom $(f^*)_1$, we get $\langle x, u^* \rangle - f(x) \leq f^*(u^*)$ and combined this gives the result.

(2) Follows immediately from (1) and $(\nabla f)_1$.

(3) Let x^* be such that

$$\forall y (\langle y - x, x^* \rangle + f(x) \leq f(y) + 2^{-j})$$

for j defined as above. This yields

$$\begin{aligned} \langle y - x, x^* - \nabla f x \rangle &= \langle y - x, x^* \rangle - \langle y - x, \nabla f x \rangle \\ &\leq f(y) - f(x) - \langle y - x, \nabla f x \rangle + 2^{-j}. \end{aligned}$$

Using Lemma 2.18, we get that for $\|y - x\| < 2^{-\omega^{\nabla f}(l, b+1)}$:

$$f(y) - f(x) - \langle y - x, \nabla f x \rangle \leq 2^{-l} \|y - x\|$$

and so $\langle y - x, x^* - \nabla f x \rangle \leq 2^{-j} + 2^{-l} \|y - x\|$ for all such y which in particular yields

$$\langle z, x^* - \nabla f x \rangle \leq 2^{-j} + 2^{-l} \|z\|$$

for all z with $\|z\| < 2^{-\omega^{\nabla f}(l, b+1)}$ given any l . For the given k , we now show that there is a z_k such that $0 < \|z_k\| \leq 1$ and

$$\langle z_k, x^* - \nabla f x \rangle \leq 2^{-(k+2)} \rightarrow \|x^* - \nabla f x\| \leq 2^{-k}.$$

To see this, suppose $\|x^* - \nabla f x\| > 2^{-k} = 2^{-(k+1)} + 2^{-(k+1)}$. Using axiom $(*)_2$, we get that there exists a z_k with $\|z_k\| \leq 1$ and such that $|\langle z_k, x^* - \nabla f x \rangle| > 2^{-(k+2)}$, which also in particular implies $\|z_k\| > 0$.

Now, using this z_k , define

$$\widehat{z}_k = 2^{-(\omega^{\nabla f}(k+3, b+1)+1)} z_k.$$

Clearly $\|\widehat{z}_k\| < 2^{-\omega^{\nabla f}(k+3, b+1)}$ and thus

$$\langle \widehat{z}_k, x^* - \nabla f x \rangle \leq 2^{-j} + 2^{-(k+3)} \|\widehat{z}_k\|$$

which yields by definition of j that

$$\begin{aligned} \langle z_k, x^* - \nabla f x \rangle &= 2^{(\omega^{\nabla f}(k+3, b+1)+1)} \langle \widehat{z}_k, x^* - \nabla f x \rangle \\ &\leq 2^{(\omega^{\nabla f}(k+3, b+1)+1)} (2^{-j} + 2^{-(k+3)} \|\widehat{z}_k\|) \\ &= 2^{(\omega^{\nabla f}(k+3, b+1)+1)} 2^{-j} + 2^{-(k+3)} 2^{(\omega^{\nabla f}(k+3, b+1)+1)} \|\widehat{z}_k\| \\ &\leq 2^{-(k+4+\omega^{\nabla f}(k+3, b+1))} 2^{(\omega^{\nabla f}(k+3, b+1)+1)} + 2^{-(k+3)} \\ &= 2^{-(k+2)} \end{aligned}$$

which implies $\|x^* - \nabla f x\| \leq 2^{-k}$ by the properties of z_k .

- (4) Let x^* be such that $f(x) + f^*(x^*) - \langle x, x^* \rangle_{X^*} \leq 2^{-j}$ with j defined as above. Then we get $f^*(x^*) \leq 2^{-j} + \langle x, x^* \rangle - f(x)$ which yields through $(f^*)_1$ that

$$\langle y, x^* \rangle - f(y) \leq 2^{-j} + \langle x, x^* \rangle - f(x)$$

for all y . This is equivalent to $\langle y - x, x^* \rangle + f(x) \leq f(y) + 2^{-j}$ for all y . Then item (3) yields the result.

- (5) This follows immediately from (3).
 (6) Suppose

$$|tf^*(x^*) + (1-t)f^*(y^*) - f^*(tx^* + (1-t)y^*)| \leq_{\mathbb{R}} 2^{-j}$$

for j as above. Then write $z^* = tx^* + (1-t)y^*$ and pick $x = \nabla f^* z^*$, i.e. $\nabla f x = z^*$ by (L). Then by item (2), the extensionality of f^* (recall Lemma 2.18) and the extensionality of $\langle \cdot, \cdot \rangle$, we get

$$\begin{aligned} 0 &= f(x) + f^*(z^*) - \langle x, z^* \rangle \\ &\geq f(x) + tf^*(x^*) + (1-t)f^*(y^*) - 2^{-j} - \langle x, z^* \rangle, \end{aligned}$$

i.e. we have

$$2^{-j} \geq t(f(x) + f^*(x^*) - \langle x, x^* \rangle) + (1-t)(f(x) + f^*(y^*) - \langle x, y^* \rangle)$$

and thus, using $t, 1-t \geq 2^{-i}$ and that $f(x) + f^*(x^*) - \langle x, x^* \rangle \geq 0$ as well as $f(x) + f^*(y^*) - \langle x, y^* \rangle \geq 0$ by the Fenchel-Young inequality (which follows from axiom $(f^*)_1$), we get

$$2^{-j} 2^i \geq f(x) + f^*(x^*) - \langle x, x^* \rangle, f(x) + f^*(y^*) - \langle x, y^* \rangle$$

By definition of j , we get

$$2^{-(k+4+\omega^{\nabla f}(k+4, F(b)+2))} \geq f(x) + f^*(x^*) - \langle x, x^* \rangle, f(x) + f^*(y^*) - \langle x, y^* \rangle.$$

Noting that $\|z^*\| \leq t\|x^*\| + (1-t)\|y^*\| < b$ and thus $\|x\| < F(b) + 1$, item (4) implies that

$$\|x^* - \nabla f x\|, \|y^* - \nabla f x\| \leq 2^{-(k+1)}$$

which yields $\|x^* - y^*\| \leq 2^{-k}$.

(7) Using item (6), note that for $t = 1/2$, we have

$$\begin{aligned} f^* \left(\frac{x^* + y^*}{2} \right) &\leq 1/2 f^*(y^*) + 1/2 f^*(x^*) - 2^{-j} \\ &= f^*(x^*) + 1/2(f^*(y^*) - f^*(x^*)) - 2^{-j} \end{aligned}$$

if $\|x^* - y^*\| > 2^{-k}$. As

$$\langle \nabla f^* w^*, z^* \rangle \leq \frac{f^*(w^* + \alpha z^*) - f^*(w^*)}{\alpha},$$

for any $\alpha > 0$ (using $(\nabla f^*)_1$), we get

$$\langle \nabla f^* x^*, y^* - x^* \rangle \leq f^*(y^*) - f^*(x^*) - 2 \cdot 2^{-j}.$$

Similarly, we get

$$\langle \nabla f^* y^*, x^* - y^* \rangle \leq f^*(x^*) - f^*(y^*) - 2 \cdot 2^{-j}$$

and this implies

$$\langle \nabla f^* y^* - \nabla f^* x^*, x^* - y^* \rangle \leq -4 \cdot 2^{-j}$$

which gives the claim. $\square$

Now, the additional axioms in $\mathcal{D}_{f,f^*}^\omega[\text{FM}]$ can be used to carry out the above proof with the roles of f and f^* exchanged. We collect this in the following lemma.

Lemma 3.5. *The system $\mathcal{D}_{f,f^*}^\omega[\text{FM}]$ proves:*

(1) *The “Fenchel-Young equality” for any subgradient u of f^* at x^* , i.e.*

$$\begin{aligned} \forall x^{*X^*}, u^X (\forall y^{*X^*} (f^*(y^*) \geq_{\mathbb{R}} f^*(x^*) + \langle u, y^* -_{X^*} x^* \rangle_{X^*}) \\ \rightarrow f^*(x^*) + f(u) =_{\mathbb{R}} \langle u, x^* \rangle_{X^*}). \end{aligned}$$

(2) *The “Fenchel-Young equality” for $\nabla f^* x$, i.e.*

$$\forall x^{*X^*} (f^*(x^*) + f(\nabla f^* x^*) =_{\mathbb{R}} \langle \nabla f^* x^*, x^* \rangle_{X^*}).$$

(3) *Approximate subgradients of f^* are close to the gradient of f^* , i.e.*

$$\begin{aligned} \forall b^0, k^0 \exists j^0 \forall x^{*X^*}, x^X (\|x^*\|_{X^*} <_{\mathbb{R}} b \\ \wedge \forall y^{*X^*} (\langle x, y^* -_{X^*} x^* \rangle_{X^*} + f^*(x^*) \leq_{\mathbb{R}} f^*(y^*) + 2^{-j}) \\ \rightarrow \|x -_X \nabla f^* x^*\|_X \leq_{\mathbb{R}} 2^{-k}), \end{aligned}$$

where in fact we can take

$$j = k + 4 + \omega^{\nabla f^*}(k + 3, b + 1).$$

(4) *The “Fenchel-Young equality” characterizes gradients of f^* , i.e.*

$$\forall x^{*X^*}, x^X (f^*(x^*) + f(x) =_{\mathbb{R}} \langle x, x^* \rangle_{X^*} \rightarrow x =_X \nabla f^* x^*),$$

where in fact it moreover holds that

$$\begin{aligned} \forall b^0, k^0 \exists j^0 \forall x^{*X^*}, x^X (\|x^*\|_{X^*} <_{\mathbb{R}} b \wedge f^*(x^*) + f(x) - \langle x, x^* \rangle_{X^*} \leq_{\mathbb{R}} 2^{-j} \\ \rightarrow \|x -_X \nabla f^* x^*\|_X \leq_{\mathbb{R}} 2^{-k}) \end{aligned}$$

where we can take

$$j = k + 4 + \omega^{\nabla f^*}(k + 3, b + 1).$$

(5) $\nabla f^* x^*$ is the unique subgradient of f^* at x^* , i.e.

$$\forall x^{*X^*}, u^X \left(\forall y^{*X^*} (f^*(y^*) \geq_{\mathbb{R}} f^*(x^*) + \langle u, y^* -_{X^*} x^* \rangle_{X^*}) \rightarrow u =_X \nabla f^* x^* \right).$$

(6) f is uniformly strictly convex on bounded subsets, i.e.

$$\begin{aligned} \forall k^0, i^0, b^0 \exists j^0 \forall x^X, y^X, t^1 (\|x\|_X, \|y\|_X <_{\mathbb{R}} b \wedge 2^{-i} \leq_{\mathbb{R}} t \leq_{\mathbb{R}} 1 - 2^{-i} \\ \wedge t f(x) + (1 - t) f(y) - f(tx +_X (1 - t)y) \leq_{\mathbb{R}} 2^{-j} \\ \rightarrow \|x -_X y\|_X \leq_{\mathbb{R}} 2^{-k}) \end{aligned}$$

where we in fact can choose

$$j = (k + 4 + \omega^{\nabla f^*}(k + 4, C(b) + 2)) + i$$

where C is a modulus for ∇f being bounded on bounded sets (which can be constructed as in Lemma 2.18).

(7) ∇f is uniformly strictly monotone on bounded subsets, i.e.

$$\begin{aligned} \forall k^0, b^0 \exists j^0 \forall x^X, y^X (\|x\|_X, \|y\|_X <_{\mathbb{R}} b \wedge \|x -_X y\|_X >_{\mathbb{R}} 2^{-k} \\ \rightarrow \langle x -_X y, \nabla f x -_{X^*} \nabla f y \rangle_{X^*} \geq_{\mathbb{R}} 2^{-j}). \end{aligned}$$

where we in fact can choose

$$j = k + 5 + \omega^{\nabla f^*}(k + 4, C(b) + 2).$$

with all other constants as in (6).

In particular, in the system $\mathcal{D}_{f,f^*}^{\omega}[\text{FM}]$ we can now formally establish some of the central properties of Bregman distances used extensively throughout the applications given in [84] and which are also crucial for the development of the theory of the relativized resolvents. We begin with the fact that $W_f(x, \nabla f(y)) = D_f(x, y)$, which arises through a simple calculation:

Lemma 3.6. *The system $\mathcal{D}_{f,f^*}^{\omega}[\text{FM}]$ proves:*

$$\forall x^X, y^X (D_f(x, y) =_{\mathbb{R}} f(x) + f^*(\nabla f y) - \langle x, \nabla f y \rangle_{X^*}).$$

Proof. By Lemma 3.4, (2), we have

$$f^*(\nabla f y) = \langle y, \nabla f y \rangle - f(y)$$

and thus

$$f(x) + f^*(\nabla f y) - \langle x, \nabla f y \rangle = f(x) - f(y) - \langle x - y, \nabla f y \rangle = D_f(x, y). \quad \square$$

The following result shows that D_f is uniformly bounded in the sense of [84], with the proof given here being a formal variant of the argument given in Remark 3.14 therein.

Lemma 3.7. *The system $\mathcal{D}_{f,f^*}^{\omega}[\text{FM}]$ proves that D_f is uniformly bounded in the sense of [84], i.e.*

$$\forall b^0, \alpha^0 \exists o^0 \forall x^X, y^X (\|x\|_X <_{\mathbb{R}} b \wedge D_f(x, y) <_{\mathbb{R}} \alpha \rightarrow \|y\|_X \leq_{\mathbb{R}} o)$$

and o can be realized by

$$o = o(\alpha, b) = F(\max\{\alpha^{f^*}(\alpha + D(b) + b), 1\} + 1)$$

where D, F are moduli of $f, \nabla f^*$ being bounded on bounded sets, respectively, and α^{f^*} is a modulus of supercoercivity for f^* as before.

Proof. First, note that $f^*(x^*) - \langle x, x^* \rangle$ is also supercoercive. For this, let $\|x\| < b$. If $\|x^*\| > \alpha^{f^*}(K + b)$, from axiom $(f^*)_3$ we derive

$$f^*(x^*) - \langle x, x^* \rangle \geq f^*(x^*) - \|x\| \|x^*\| \geq (K + b - \|x\|) \|x^*\| \geq K \|x^*\|.$$

Now, we have

$$f^*(\nabla f y) - \langle x, \nabla f y \rangle = D_f(x, y) - f(x) < \alpha + D(b)$$

using the above Lemma 3.6. Therefore, we derive

$$\|\nabla f y\| \leq \max\{\alpha^{f^*}(\alpha + D(b) + b), 1\}$$

and thus we get $\|y\| = \|\nabla f^* \nabla f y\| \leq F(\max\{\alpha^{f^*}(\alpha + D(b) + b), 1\} + 1)$. $\square$

3.2. Monotone operators and their relativized resolvents. We now move to the central new objects, the monotone operators in the sense of Browder and their relativized resolvents. Initially, generic set-valued operators of the form $A : X \rightarrow 2^{X^*}$ are, in similarity to [76], modeled via a constant for their characteristic function. In the context of the system $\mathcal{D}_{f,f^*}^\omega[\text{FM}]$, we in that way add a constant χ_A of type $0(X^*)(X)$ and write $x^* \in Ax$, $(x, x^*) \in A$ or $(x, x^*) \in \text{gra}A$ for $\chi_A x x^* =_0 0$. The first natural axiom is

$$(I)^* \quad \forall x^X, x^{*X^*} (\chi_A x x^* \leq_0 1)$$

which witnesses that χ_A is a characteristic function.

Also, the treatment of the resolvent is conceptually similar to that work. For this, let A be monotone (in the sense of Browder, recall Section 2.1) and recall Definition 2.11 for the f -resolvents of such monotone operators, whereby $\text{Res}_{\gamma A}^f : X \rightarrow 2^X$ is defined by

$$\text{Res}_{\gamma A}^f x := ((\nabla f + \gamma A)^{-1} \circ \nabla f)(x)$$

for any $x \in X$ and $\gamma > 0$. It follows by our assumptions on f (recall Assumption 3.1) and Proposition 2.13 that this map is single-valued, satisfies $\text{Fix}(\text{Res}_{\gamma A}^f) = A^{-1}0$ (noting that $\text{dom}f = X$) and that it is Bregman firmly nonexpansive.

So, for treating an operator A with total relativized resolvents, we add a constant Res_A^f of type $X(X)(1)$ and write $\text{Res}_{\gamma A}^f$ for $\text{Res}_A^f \gamma$. The natural axiom for the resolvent can now be derived similar as to [76]: If seen as a set-valued operator, the resolvent satisfies

$$\begin{aligned} p \in \text{Res}_{\gamma A}^f x &\Leftrightarrow p \in (\nabla f + \gamma A)^{-1} \nabla f(x) \\ &\Leftrightarrow \nabla f(x) \in \nabla f(p) + \gamma A p \\ &\Leftrightarrow \gamma^{-1} (\nabla f(x) - \nabla f(p)) \in A p. \end{aligned}$$

This naturally leads us to consider the axiom scheme

$$(II)^* \quad \forall \gamma^1, x^X \left(\gamma >_{\mathbb{R}} 0 \rightarrow \gamma^{-1} (\nabla f x -_{X^*} \nabla f(\text{Res}_{\gamma A}^f x)) \in A(\text{Res}_{\gamma A}^f x) \right)$$

in similarity to axiom (II) considered in [76], stating that $\text{Res}_{\gamma A}^f$ is a selection map of the (a priori set-valued) relativized resolvent. In fact, this axiom (together with the axiom specified previously and the two axioms specified further below) will be sufficient for developing all the main parts of the theory of relativized resolvents (e.g. immediately entailing the uniqueness and Bregman firm nonexpansivity of the above selection map) as we will later discuss.

Remark 3.8. As in the context of the systems from [76], note that also here, the above axiom $(II)^*$ is actually an abbreviation for the following sentence where the dependence of γ^{-1} on a lower bound of γ is made explicit:

$$\forall \gamma^1, x^X, k^0 \left(\gamma >_{\mathbb{R}} 2^{-k} \rightarrow (\gamma)_k^{-1} (\nabla f x -_{X^*} \nabla f(\text{Res}_{\gamma A}^f x)) \in A(\text{Res}_{\gamma A}^f x) \right).$$

However, also similar to [76], we will still in general employ the above shorthand style of writing where the parameter k is omitted as in this paper, the context will always make it clear how a given statement in that style has to be expanded to yield a proper formal variant so that no issues arise.

Also the monotonicity of A is easily specified by a universal axiom:

$$(III)^* \quad \forall x^X, y^X, x^{*X^*}, y^{*X^*} ((x, x^*), (y, y^*) \in A \rightarrow \langle x -_X y, x^* -_{X^*} y^* \rangle_{X^*} \geq_{\mathbb{R}} 0).$$

Lastly, all uses of the relativized resolvent in proof mining applications presented so far (see in particular [84]) are made in the context of the assumption that $A^{-1}0 \neq \emptyset$ and we will also assume this here as it in particular will allow us to majorize the resolvent rather immediately (as already mentioned before in the context of Assumption 3.2). For this, we add a constant p_X of type X together with a corresponding axiom stating that p_X is a zero of A :

$$(IV)^* \quad 0_{X^*} \in Ap_X.$$

This leads us to the following system:

Definition 3.9. The system $\mathcal{B}^\omega$ is defined as the extension of the system $\mathcal{D}_{f,f^*}^\omega[\text{FM}]$ with the above constants and corresponding axioms $(I)^*$ - $(IV)^*$.

Remark 3.10. Alternatively to adding a precise witness p_X for a zero of the operator A , we could also have added only a constant b_z of type 0 representing a norm upper bound on such a zero, which is the only numerical data on which bounds extracted through the metatheorem later depend, together with the corresponding axiom $\exists p \leq_X b_z 1_X (0_{X^*} \in Ap)$ which is of type Δ and hence admissible in our systems.

Now, in similarity to the systems from [76], also $\mathcal{B}^\omega$ is sufficient for formalizing the first main aspects of the theory of monotone operators in Banach spaces and their f -resolvents as the following proposition shows.

Lemma 3.11. *The system $\mathcal{B}^\omega$ proves:*

(1) $\text{Res}_{\gamma A}^f$ is unique for any $\gamma > 0$, i.e.

$$\forall \gamma^1, p^X, x^X \left(\gamma >_{\mathbb{R}} 0 \wedge \gamma^{-1} (\nabla f x -_{X^*} \nabla f p) \in Ap \rightarrow p =_X \text{Res}_{\gamma A}^f x \right).$$

(2) $\text{Res}_{\gamma A}^f$ is Bregman firmly nonexpansive for any $\gamma > 0$, i.e.

$$\begin{aligned} \forall \gamma^1, x^X, y^X (\gamma >_{\mathbb{R}} 0 \rightarrow \langle \text{Res}_{\gamma A}^f x -_X \text{Res}_{\gamma A}^f y, \nabla f \text{Res}_{\gamma A}^f x -_{X^*} \nabla f \text{Res}_{\gamma A}^f y \rangle_{X^*} \\ \leq_{\mathbb{R}} \langle \text{Res}_{\gamma A}^f x -_X \text{Res}_{\gamma A}^f y, \nabla f x -_{X^*} \nabla f y \rangle_{X^*}). \end{aligned}$$

(3) $\text{Res}_{\gamma A}^f$ satisfies the alternative notion of Bregman firm nonexpansivity for any $\gamma > 0$, i.e.

$$\begin{aligned} \forall \gamma^1, x^X, y^X (\gamma >_{\mathbb{R}} 0 \rightarrow D_f(\text{Res}_{\gamma A}^f x, \text{Res}_{\gamma A}^f y) + D_f(\text{Res}_{\gamma A}^f y, \text{Res}_{\gamma A}^f x) \\ \leq_{\mathbb{R}} D_f(\text{Res}_{\gamma A}^f x, y) + D_f(\text{Res}_{\gamma A}^f y, x) - D_f(\text{Res}_{\gamma A}^f x, x) - D_f(\text{Res}_{\gamma A}^f y, y)). \end{aligned}$$

(4) $A^{-1}0 \subseteq \text{Fix}(\text{Res}_{\gamma A}^f)$ for any $\gamma > 0$, i.e.

$$\forall p^X, \gamma^1 \left(\gamma >_{\mathbb{R}} 0 \wedge 0 \in Ap \rightarrow p =_X \text{Res}_{\gamma A}^f p \right).$$

The above result represents a formal variant of parts of Proposition 2.13, and as such the proofs all directly follow the standard arguments from the literature (which can e.g. be found in [6]).

Proof of Lemma 3.11. (1) Suppose that $\gamma > 0$ and that $\gamma^{-1}(\nabla f x - \nabla f p) \in Ap$. Axiom $(II)^*$ gives $\gamma^{-1}(\nabla f x - \nabla f \text{Res}_{\gamma A}^f x) \in A(\text{Res}_{\gamma A}^f x)$. Axiom $(III)^*$ then implies that

$$\begin{aligned} 0 &\leq \langle \text{Res}_{\gamma A}^f x - p, \gamma^{-1}(\nabla f x - \nabla f \text{Res}_{\gamma A}^f x) - \gamma^{-1}(\nabla f x - \nabla f p) \rangle \\ &= \langle \text{Res}_{\gamma A}^f x - p, \gamma^{-1}(\nabla f p - \nabla f \text{Res}_{\gamma A}^f x) \rangle \end{aligned}$$

where we have used extensionality of $\langle \cdot, \cdot \rangle$ and of the arithmetical operations in X^* . In particular, since $\gamma^{-1} > 0$ as $\gamma > 0$, we get that

$$\langle \text{Res}_{\gamma A}^f x - p, \nabla f \text{Res}_{\gamma A}^f x - \nabla f p \rangle \leq 0.$$

Thus, as ∇f is provably strictly monotone (Lemma 3.5), we get $\|\text{Res}_{\gamma A}^f x - p\| = 0$, i.e.

$$\text{Res}_{\gamma A}^f x = p.$$

(2) Let $\gamma > 0$. Axiom $(II)^*$ gives

$$\gamma^{-1}(\nabla f x - \nabla f \text{Res}_{\gamma A}^f x) \in A(\text{Res}_{\gamma A}^f x) \text{ and } \gamma^{-1}(\nabla f y - \nabla f \text{Res}_{\gamma A}^f y) \in A(\text{Res}_{\gamma A}^f y).$$

Axiom $(III)^*$ and $\gamma^{-1} > 0$ gives

$$\langle \text{Res}_{\gamma A}^f x - \text{Res}_{\gamma A}^f y, \nabla f x - \nabla f y - (\nabla f \text{Res}_{\gamma A}^f x - \nabla f \text{Res}_{\gamma A}^f y) \rangle \geq 0$$

which implies

$$\langle \text{Res}_{\gamma A}^f x - \text{Res}_{\gamma A}^f y, \nabla f x - \nabla f y \rangle \geq \langle \text{Res}_{\gamma A}^f x - \text{Res}_{\gamma A}^f y, \nabla f \text{Res}_{\gamma A}^f x - \nabla f \text{Res}_{\gamma A}^f y \rangle.$$

(3) By the provability of the three-point identity for D_f (Lemma 2.19), we get

$$\begin{aligned} &\langle \text{Res}_{\gamma A}^f x - \text{Res}_{\gamma A}^f y, \nabla f \text{Res}_{\gamma A}^f x - \nabla f \text{Res}_{\gamma A}^f y \rangle \\ &= D_f(\text{Res}_{\gamma A}^f x, \text{Res}_{\gamma A}^f y) + D_f(\text{Res}_{\gamma A}^f y, \text{Res}_{\gamma A}^f x) - D_f(\text{Res}_{\gamma A}^f x, \text{Res}_{\gamma A}^f x) \\ &= D_f(\text{Res}_{\gamma A}^f x, \text{Res}_{\gamma A}^f y) + D_f(\text{Res}_{\gamma A}^f y, \text{Res}_{\gamma A}^f x). \end{aligned}$$

Further, by the provability of the four-point identity for D_f (Lemma 2.19), we get

$$\begin{aligned} &\langle \text{Res}_{\gamma A}^f x - \text{Res}_{\gamma A}^f y, \nabla f x - \nabla f y \rangle \\ &= D_f(\text{Res}_{\gamma A}^f x, y) - D_f(\text{Res}_{\gamma A}^f x, x) - D_f(\text{Res}_{\gamma A}^f y, y) + D_f(\text{Res}_{\gamma A}^f y, x). \end{aligned}$$

Thus, using item (2), we get the claimed inequality.

(4) Let p be such that $0 \in Ap$. Then provably with the only assumption being $\gamma > 0$, we have $\gamma^{-1}(\nabla f p - \nabla f p) = 0$ and thus, using Σ_1 -ER (recall Remark 2.15), we have that $0 \in Ap$ implies

$$\gamma^{-1}(\nabla f p - \nabla f p) \in Ap.$$

Using item (1), we get $p = \text{Res}_{\gamma A}^f p$. □

Also the boundedness and continuity properties of maps that are Bregman firmly nonexpansive, already emphasized in [84] as crucial for the practice of proof mining for these objects, can now be formally replicated in the context of the system $\mathcal{B}^\omega$ (where we here formulate these properties just for the resolvents):

Lemma 3.12. *The system $\mathcal{B}^\omega$ proves:*

(1) $\text{Res}_{\gamma A}^f$ is bounded on bounded sets for any $\gamma > 0$, i.e.

$$\forall \gamma^1, b^0 \exists e^0 \forall x^X \left(\gamma >_{\mathbb{R}} 0 \wedge \|p_X\|_X, \|x\|_X <_{\mathbb{R}} b \rightarrow \left\| \text{Res}_{\gamma A}^f x \right\|_X \leq_{\mathbb{R}} e \right),$$

where in fact one can choose

$$e = E(b) = o(2D(b) + 2bC(b), b)$$

where C, D are moduli witnessing that $\nabla f, f$ are bounded on bounded sets, respectively, and o is defined as in Lemma 3.7.

(2) $\text{Res}_{\gamma_A}^f$ is uniformly continuous on bounded sets for any $\gamma > 0$, i.e.

$$\begin{aligned} & \forall \gamma^1, k^0, b^0 \exists j^0 \forall x^X, y^X (\gamma >_{\mathbb{R}} 0 \wedge \|p_X\|_X, \|x\|_X, \|y\|_X <_{\mathbb{R}} b \\ & \wedge \|x -_X y\|_X <_{\mathbb{R}} 2^{-j} \rightarrow \left\| \text{Res}_{\gamma_A}^f x -_X \text{Res}_{\gamma_A}^f y \right\|_X \leq_{\mathbb{R}} 2^{-k}) \end{aligned}$$

where in fact one can choose

$$j = \varpi(k, b) = \omega^{\nabla f}(\widehat{k} + 1 + E(b), b)$$

for $\widehat{k} = k + 5 + \omega^{\nabla f^*}(k + 4, C(b) + 2)$ with C being a modulus witnessing that ∇f is bounded on bounded sets and where E is defined as in (1).

Notice that therefore the system $\mathcal{B}^\omega$ proves that $\text{Res}_{\gamma_A}^f$ is extensional. The proofs for the above two items in particular are formal variants of arguments given for Bregman firmly nonexpansive mappings in [84] (see Lemma 4.6 and Lemma 4.14 therein).

Proof of Lemma 3.12. For item (1), note that by Lemma 3.11, (3) and (4), and with $p = p_X$ from axiom (IV)*, we have (using the extensionality of D_f which follows from that of $f, \nabla f$ and $\langle \cdot, \cdot \rangle$):

$$\begin{aligned} D_f(\text{Res}_{\gamma_A}^f x, p) + D_f(p, \text{Res}_{\gamma_A}^f x) & \leq D_f(\text{Res}_{\gamma_A}^f x, p) + D_f(p, x) \\ & \quad - D_f(\text{Res}_{\gamma_A}^f x, x) - D_f(p, p) \\ & \leq D_f(\text{Res}_{\gamma_A}^f x, p) + D_f(p, x) \end{aligned}$$

and thus

$$D_f(p, \text{Res}_{\gamma_A}^f x) \leq D_f(p, x) < 2D(b) + 2bC(b).$$

Thus, by Lemma 3.7, we get

$$\left\| \text{Res}_{\gamma_A}^f x \right\| \leq o(2D(b) + 2bC(b), b).$$

For item (2), by Lemma 3.11, (2), we have

$$\begin{aligned} & \langle \text{Res}_{\gamma_A}^f x - \text{Res}_{\gamma_A}^f y, \nabla f \text{Res}_{\gamma_A}^f x - \nabla f \text{Res}_{\gamma_A}^f y \rangle \\ & \leq \langle \text{Res}_{\gamma_A}^f x - \text{Res}_{\gamma_A}^f y, \nabla f x - \nabla f y \rangle \\ & \leq \left\| \text{Res}_{\gamma_A}^f x - \text{Res}_{\gamma_A}^f y \right\| \left\| \nabla f x - \nabla f y \right\| \\ & \leq 2E(b) \left\| \nabla f x - \nabla f y \right\|, \end{aligned}$$

using also additionally the above item (1). So, for $\|x - y\| < 2^{-j}$ with the j defined above, we have

$$\left\| \nabla f x - \nabla f y \right\| \leq 2^{-(\widehat{k} + 1 + E(b))}$$

and thus

$$\langle \text{Res}_{\gamma_A}^f x - \text{Res}_{\gamma_A}^f y, \nabla f \text{Res}_{\gamma_A}^f x - \nabla f \text{Res}_{\gamma_A}^f y \rangle < 2^{-\widehat{k}}.$$

Thus by Lemma 3.5, (7), we get

$$\left\| \text{Res}_{\gamma_A}^f x - \text{Res}_{\gamma_A}^f y \right\| \leq 2^{-k}.$$

□

4. MAXIMALITY AND EXTENSIONALITY

Naturally, the system $\mathcal{B}^\omega$ cannot prove the extensionality of A since we also deal with potentially discontinuous operators A semantically. As mentioned in the introduction, a central theoretical result from [76] connects the extensionality of A with the maximality statement for A in the setting of monotone operators on Hilbert spaces. The present section now extends this equivalence to monotone operators over Banach spaces. In particular, by that equivalence, this maximality statement cannot be provable in the system either. Further, we extend this collection of equivalences between extensionality principles and mathematical principles from operator theory with ones of a new type, specifically phrased for the zero set of the operator (which retroactively also apply to the systems from [76]).

For the practice of proof mining, these results seems dire at first: How should the systems described in [76], and by extension the systems of the present paper, be applicable to proofs from monotone operator theory when most analytical results from that area commonly rely on maximality assumptions? These issues get amplified by the fact that it was shown in [82] that the extensionality principle for set-valued operators cannot be added to a system such as $\mathcal{B}^\omega$ meanwhile maintaining a metatheorem in the style of proof mining (as e.g. established later), so that in particular these metatheorems are in a sense permanently precluded from treating proofs involving the corresponding maximality principle.

These circumstances are practically mitigated by the empirical fact, observed by inspecting large parts of the associated literature, that a maximality assumption is commonly only used to guarantee the existence of total resolvent. Moreover, it is generally already “only” an intensional treatment of these total resolvents as given above which is sufficient for treating most arguments. These empirical observations in particular underlay and informed the design choices of the systems from [76] as well as the present ones, and are in particular supported by the many case studies carried out therein since (recall the references in the introduction). Indeed, essentially none of these case studied use the extensionality of the operator A in any way, and if a proof really requires such a principle, it is generally only a suitable fragment thereof, with a range these allowing for a quantitative treatment in the context of the metatheorems as illustrated in [82].

Hence, while commonly not a direct obstacle in concrete extractions, these equivalences between the extensionality principle and the maximal monotonicity of the operator A are still important theoretical hardness results, showing the limitations of the applicability of the given systems and their underlying approach to set-valued operators. The more of these equivalences are available, the clearer the limits of these systems then become, motivating a general kind of program of *reverse mathematics for extensionality principles* as relevant to proof mining, where the results from [76] and [82], but also of the present section, can hence be seen as initial results therein.

We now begin with the first equivalence of extensionality and maximal monotonicity.

Theorem 4.1. *Over $\mathcal{B}^\omega$, the following are equivalent:*

(1) *Extensionality of A , i.e.*

$$\forall x^X, x^{*X*}, y^X, y^{*X*} (x =_X y \wedge x^* =_{X^*} y^* \wedge x^* \in Ax \rightarrow y^* \in Ay).$$

(2) *The strong resolvent axiom, i.e.*

$$\forall x^X, p^X, \gamma^1 \left(\gamma >_{\mathbb{R}} 0 \wedge p =_X \text{Res}_{\gamma A}^f x \rightarrow \gamma^{-1}(\nabla f x -_{X^*} \nabla f p) \in Ap \right).$$

(3) *Maximal monotonicity of A , i.e.*

$$\forall x^X, x^{*X^*} \left(\forall y^X, y^{*X^*} (y^* \in Ay \rightarrow \langle x -_X y, x^* -_{X^*} y^* \rangle_{X^*} \geq_{\mathbb{R}} 0) \rightarrow x^* \in Ax \right).$$

Proof. For the direction (1) $\Rightarrow$ (3), let x, x^* be such that

$$\langle x - y, x^* - y^* \rangle \geq 0 \text{ for all } (y, y^*) \in A.$$

We consider $z = \nabla f^*(x^* + \nabla f x)$. Then

$$1^{-1}(\nabla f z - \nabla f \text{Res}_A^f z) \in A(\text{Res}_A^f z).$$

by axiom (II)*. Thus by the assumption on x, x^* , axiom (L) and the extensionality of $\langle \cdot, \cdot \rangle$ we get

$$\begin{aligned} 0 &\leq \langle x - \text{Res}_A^f z, x^* - (\nabla f z - \nabla f \text{Res}_A^f z) \rangle \\ &= \langle x - \text{Res}_A^f z, \nabla f \text{Res}_A^f z - \nabla f x \rangle \end{aligned}$$

which is equivalent to

$$\langle x - \text{Res}_A^f z, \nabla f x - \nabla f \text{Res}_A^f z \rangle \leq 0$$

and this yields $x = \text{Res}_A^f z$ as ∇f is (provably) strictly monotone. Further, we have

$$1^{-1}(\nabla f z - \nabla f \text{Res}_A^f z) = x^* + \nabla f x - \nabla f x = x^*$$

using (L) and the extensionality of ∇f and thus the extensionality of A yields $x^* \in Ax$.

For the direction (3) $\Rightarrow$ (2), assume that $\gamma > 0$ and $p = \text{Res}_{\gamma A}^f x$. Then at first

$$\gamma^{-1}(\nabla f x - \nabla f \text{Res}_{\gamma A}^f x) \in A(\text{Res}_{\gamma A}^f x)$$

by axiom (II)*. By monotonicity (axiom (III)* together with the extensionality of $\langle \cdot, \cdot \rangle$ and ∇f , we get

$$\forall (y, y^*) \in A(\langle p - y, \gamma^{-1}(\nabla f x - \nabla f p) - y^* \rangle \geq 0).$$

By (3), we get

$$\gamma^{-1}(\nabla f x - \nabla f p) \in A(p).$$

For (2) $\Rightarrow$ (1), let $x = y$ and $x^* = y^*$ with $x^* \in Ax$. Define

$$z = \nabla f^*(y^* + \nabla f y).$$

By (II)*, we get

$$1^{-1}(\nabla f z - \nabla f \text{Res}_A^f z) \in A(\text{Res}_A^f z).$$

Axiom (III)* together with the extensionality of $\langle \cdot, \cdot \rangle$ and ∇f^* as well as using (L) yields

$$\begin{aligned} 0 &\leq \langle x - \text{Res}_A^f z, x^* - (\nabla f z - \nabla f \text{Res}_A^f z) \rangle \\ &= \langle y - \text{Res}_A^f z, \nabla f \text{Res}_A^f z - \nabla f y \rangle \end{aligned}$$

and this is equivalent to

$$\langle y - \text{Res}_A^f z, \nabla f y - \nabla f \text{Res}_A^f z \rangle \leq 0$$

which yields $y = \text{Res}_A^f z$ by provable strict monotonicity of ∇f . Using (2), we have

$$1^{-1}(\nabla f z - \nabla f y) \in Ay$$

which yields by the quantifier-free extensionality rule that $y^* \in Ay$ as $1^{-1}(\nabla f z - \nabla f y) = y^*$ holds without any additional assumptions. $\square$

Similar to [76] however (see Theorem 3.2 therein), the system $\mathcal{B}^\omega$ does support a weakened, so-called intensional, maximality principle as in the following theorem:

Theorem 4.2. *The system $\mathcal{B}^\omega$ proves the following intensional maximality principle:*

$$\begin{aligned} & \forall x^X, x^{*X*} (\forall y^X, y^{*X*} (y^* \in Ay \rightarrow \langle x -_X y, x^* -_{X*} y^* \rangle_{X*} \geq_{\mathbb{R}} 0) \\ & \rightarrow \exists x'^X, x'^{*X*} (x =_X x' \wedge x^* =_{X*} x'^* \wedge x'^* \in Ax')). \end{aligned}$$

Proof. As in the proof of the direction (1) $\Rightarrow$ (3) from the above Theorem 4.1, we get that

$$1^{-1}(\nabla f z - \nabla f \text{Res}_A^f z) \in A(\text{Res}_A^f z)$$

together with $x = \text{Res}_A^f z$ and $1^{-1}(\nabla f z - \nabla f \text{Res}_A^f z) = x^*$ for $z = \nabla f^*(x^* + \nabla f x)$ without any use of extensionality. This gives the claim by setting $x' = \text{Res}_A^f z$ and $x'^* = 1^{-1}(\nabla f z - \nabla f \text{Res}_A^f z)$. $\square$

We end this section by presenting a result analogous to Theorem 4.1 but phrased specifically for the zero set of the operator A . At first, recall Lemma 3.11, (4), by which the system $\mathcal{B}^\omega$ proves

$$\forall x^X \left(0 \in Ax \rightarrow \forall \gamma^1 \left(\gamma >_{\mathbb{R}} 0 \rightarrow \text{Res}_{\gamma A}^f x =_X x \right) \right).$$

Further, we even provably have

$$\forall x^X, z^{X*} \left(z \in Ax \wedge z =_{X*} 0 \rightarrow \forall \gamma^1 \left(\gamma >_{\mathbb{R}} 0 \rightarrow \text{Res}_{\gamma A}^f x =_X x \right) \right).$$

To see this, note that by the quantifier-free extensionality rule we have from $z \in Ax$ that $\gamma^{-1}(\nabla f \nabla f^*(\gamma z + \nabla f x) - \nabla f x) \in Ax$ and thus

$$x =_X \text{Res}_{\gamma A}^f(\nabla f^*(\gamma z + \nabla f x)) =_X \text{Res}_{\gamma A}^f x$$

using uniqueness and extensionality of the resolvent as well as $z = 0$.

As we will see now, the converse assertions are connected to the extensionality of the set of zeros of A .

Theorem 4.3. *Over $\mathcal{B}^\omega$, the following are equivalent:*

- (1) $\forall x^X, z^{X*} \left(\text{Res}_A^f x =_X x \wedge z =_{X*} 0 \rightarrow z \in Ax \right),$
- (2) $\forall x^X, z^{X*} \left(\forall \gamma^1 \left(\gamma >_{\mathbb{R}} 0 \rightarrow \text{Res}_{\gamma A}^f x =_X x \right) \wedge z =_{X*} 0 \rightarrow z \in Ax \right),$
- (3) $\forall x^X, y^X, z^{X*}, z'^{X*} (x =_X y \wedge z =_{X*} z' =_{X*} 0 \wedge z \in Ax \rightarrow z' \in Ay).$

Proof. The implication from (1) to (2) is clear. For (2) $\Rightarrow$ (3), let $x = y$ and let $z = z' = 0$ with $z \in Ax$. Then using (2) and the provability of

$$\forall x^X, z^{X*} \left(z \in Ax \wedge z =_{X*} 0 \rightarrow \forall \gamma^1 \left(\gamma >_{\mathbb{R}} 0 \rightarrow \text{Res}_{\gamma A}^f x =_X x \right) \right)$$

as established above, we get

$$\begin{aligned} z \in Ax & \leftrightarrow \forall \gamma \left(\gamma > 0 \rightarrow \text{Res}_{\gamma A}^f x = x \right) \\ & \leftrightarrow \forall \gamma \left(\gamma > 0 \rightarrow \text{Res}_{\gamma A}^f y = y \right) \\ & \leftrightarrow z' \in Ay \end{aligned}$$

by the extensionality of $\text{Res}_{\gamma A}^f$ (recall Lemma 3.12). Lastly, for (3) $\Rightarrow$ (1), assume that $\text{Res}_A^f x = x$ and $z = 0$. Then by axiom (II)*, we get

$$1^{-1}(\nabla f x - \nabla f(\text{Res}_A^f x)) \in A(\text{Res}_A^f x).$$

By (3) and $1^{-1}(\nabla f x - \nabla f(\text{Res}_A^f x)) = 0 = z$, we get $z \in Ax$. $\square$

Further, as we will see now, this form of the extensionality of the zero set of A is even equivalent to a corresponding fragment of the maximality principle:

Theorem 4.4. *Over $\mathcal{B}^\omega$, the following are equivalent:*

$$(1) \begin{cases} \forall x^X, y^X, z^{*X^*}, z'^{*X^*} (x =_X y \\ \quad \wedge z^* =_{X^*} z'^* =_{X^*} 0 \wedge z^* \in Ax \rightarrow z'^* \in Ay), \end{cases}$$

$$(2) \begin{cases} \forall x^X, z^{*X^*} (z^* =_{X^*} 0 \\ \quad \wedge \forall y^X, y^{*X^*} (y^* \in Ay \rightarrow \langle x -_X y, -_{X^*} y^* \rangle_{X^*} \geq_{\mathbb{R}} 0) \rightarrow z^* \in Ax). \end{cases}$$

Proof. For the direction (1) $\Rightarrow$ (2), assume $\forall (y, y^*) \in A (\langle x - y, -y^* \rangle \geq 0)$. By axiom (II)*, we have $1^{-1}(\nabla f x - \nabla f(\text{Res}_A^f x)) \in A(\text{Res}_A^f x)$. Therefore, we have

$$\langle x - \text{Res}_A^f x, \nabla f x - \nabla f(\text{Res}_A^f x) \rangle \leq 0$$

and since ∇f is strictly monotone (recall Lemma 3.5), we get $x = \text{Res}_A^f x$ as well as $1^{-1}(\nabla f x - \nabla f(\text{Res}_A^f x)) = 0$. Thus (1) yields $z^* \in Ax$ for any $z^* = 0$.

Conversely, for (2) $\Rightarrow$ (1), assume that $x = y$ and $z^* = z'^* = 0$ as well as $z^* \in Ax$. Then by (III)*, we have for all $(y, y^*) \in A$:

$$\langle x - y, -y^* \rangle = \langle x - y, z^* - y^* \rangle \geq 0.$$

Thus (2) yields that $z'^* \in Ay$. □

Naturally, also the extensionality of the zero set of A as formulated above, and hence the other principles equivalent to it, cannot be provable in $\mathcal{B}^\omega$ by virtue of the upcoming bound extraction theorems. However, akin to Theorem 4.2 above, we have the provability of the following intensional representation of the inclusion of the fixed point set of the resolvent in the zero set of the operator:

Theorem 4.5. *The system $\mathcal{B}^\omega$ proves the following principle:*

$$\forall x^X \left(\text{Res}_A^f x =_X x \rightarrow \exists x'^X, z^{X^*} (x' =_X x \wedge z =_{X^*} 0 \wedge z \in Ax') \right).$$

Proof. By axiom (II)*, we get that $1^{-1}(\nabla f x - \nabla f \text{Res}_A^f x) \in A(\text{Res}_A^f x)$. This gives the claim by setting $x' = \text{Res}_A^f x$ and $z = 1^{-1}(\nabla f x - \nabla f \text{Res}_A^f x)$ as by $\text{Res}_A^f x = x$, we have $x' = x$ as well as $z = 0$ using the extensionality of ∇f . □

Remark 4.6. Similar results as the above for the zero set of the operator A naturally also hold for the systems and operators considered in [76].

5. A BOUND EXTRACTION THEOREM

We now prove the central results of this paper, the bound extraction theorem for the theory $\mathcal{B}^\omega$. As discussed before, this bound extraction theorem arises from and extends those for $\mathcal{D}^\omega$ and its extensions proven in [77] as well as those proven in [76] for the systems dealing with monotone operators in Hilbert spaces. In that vein, we keep the proofs short and only briefly discuss the key ingredients, it being implied that all considerations thus abbreviated and that regard the dual space can be made similar to [77] and those that regard the operator A can be made similar to [76].

The high-level outline of the proof of the metatheorem for $\mathcal{B}^\omega$ given here is now very much standard, following that of essentially all other such metatheorems: At first, one extracts realizers from proofs of (essentially) $\forall\exists$ -theorems using a combination of Gödel's functional interpretation with a negative translation. The resulting realizers have types from T^{X, X^*} and

we then use majorizability to construct bounds with types from T for these realizers, depending only on majorants of the parameters, which are validated in a model based on $\mathcal{M}^{\omega, X, X^*}$, the structure of all strongly majorizable functionals defined using a Banach space X and its dual X^* . In a final step, we can then recover to the truth in a model based on the usual full set-theoretic structure $\mathcal{S}^{\omega, X, X^*}$ if the types occurring in the axioms and the theorem are “low enough” (later called admissible).

We now discuss the first major ingredient of this approach, the functional interpretation and its corresponding soundness result. Originally introduced in Gödel’s seminal paper [27], the functional interpretation assigns to every formula $F(\underline{a})$ another formula $F^D(\underline{a}) = \exists \underline{x} \forall \underline{y} F_D(\underline{x}, \underline{y}, \underline{a})$, where $F_D(\underline{x}, \underline{y}, \underline{a})$ is quantifier-free (and the length of the tuples $\underline{x}, \underline{y}$ and their types depend on the structure of F), via the following recursive definition (where we for simplicity omit the free variables $\underline{a}$):

Definition 5.1 ([27]). The Dialectica interpretation $F^D = \exists \underline{x} \forall \underline{y} F_D(\underline{x}, \underline{y})$ of a formula F in the language of $\mathcal{A}^\omega[X, \|\cdot\|]$ (and its extensions) is defined via the following recursion on the structure of the formula:

(1) $F^D := F_D := F$ for F being a prime formula.

If $F^D = \exists \underline{x} \forall \underline{y} F_D(\underline{x}, \underline{y})$ and $G^D = \exists \underline{u} \forall \underline{v} G_D(\underline{u}, \underline{v})$, we set

(2) $(F \wedge G)^D := \exists \underline{x}, \underline{u} \forall \underline{y}, \underline{v} (F \wedge G)_D$

where $(F \wedge G)_D(\underline{x}, \underline{u}, \underline{y}, \underline{v}) := F_D(\underline{x}, \underline{y}) \wedge G_D(\underline{u}, \underline{v})$,

(3) $(F \vee G)^D := \exists z^0, \underline{x}, \underline{u} \forall \underline{y}, \underline{v} (F \vee G)_D$

where $(F \vee G)_D(z^0, \underline{x}, \underline{u}, \underline{y}, \underline{v}) := (z = 0 \rightarrow F_D(\underline{x}, \underline{y})) \wedge (z \neq 0 \rightarrow G_D(\underline{u}, \underline{v}))$,

(4) $(F \rightarrow G)^D := \exists \underline{U}, \underline{Y} \forall \underline{x}, \underline{v} (F \rightarrow G)_D$

where $(F \rightarrow G)_D(\underline{U}, \underline{Y}, \underline{x}, \underline{v}) := F_D(\underline{x}, \underline{Y} \underline{x} \underline{v}) \rightarrow G_D(\underline{U} \underline{x}, \underline{v})$,

(5) $(\exists z^\tau F(z))^D := \exists z, \underline{x} \forall \underline{y} (\exists z^\tau F(z))_D$

where $(\exists z^\tau F(z))_D(z, \underline{x}, \underline{y}) := F_D(\underline{x}, \underline{y}, z)$,

(6) $(\forall z^\tau F(z))^D := \exists \underline{X} \forall z, \underline{y} (\forall z^\tau F(z))_D$

where $(\forall z^\tau F(z))_D(\underline{X}, z, \underline{y}) := F_D(\underline{X} z, \underline{y}, z)$.

As mentioned before, we will consider this interpretation in combination with a negative translation that serves to translate from a classical system to its intuitionistic counterpart over which the Dialectica interpretation can then be applied. For that, we fix the following negative translation first defined by Kuroda:

Definition 5.2 ([59]). The negative translation of F is defined by $F' := \neg \neg F^*$ where F^* is defined by the following recursion on the structure of F :

(1) $F^* := F$ for prime F ;

(2) $(F \circ G)^* := F^* \circ G^*$ for $\circ \in \{\wedge, \vee, \rightarrow\}$;

(3) $(\exists x^\tau F)^* := \exists x^\tau F^*$;

(4) $(\forall x^\tau F)^* := \forall x^\tau \neg \neg F^*$.

The main result for the combination of both of these interpretations is then the following soundness result:

Lemma 5.3 (essentially [39]). *Let $\mathcal{P}$ be a set of universal sentences and let $F(\underline{a})$ be an arbitrary formula in the language of $\mathcal{A}^\omega[X, \|\cdot\|]$, the latter with only the variables $\underline{a}$ free. Then the rule*

$$\left\{ \begin{array}{l} \mathcal{A}^\omega[X, \|\cdot\|] + \mathcal{P} \vdash F(\underline{a}) \Rightarrow \\ \mathcal{A}^\omega[X, \|\cdot\|]^- + (\text{BR}) + \mathcal{P} \vdash \forall \underline{a}, \underline{y} (F')_D(\underline{t} \underline{a}, \underline{y}, \underline{a}) \end{array} \right.$$

holds where $\underline{t}$ is a tuple of closed terms of $\mathcal{A}^\omega[X, \|\cdot\|]^- + (\text{BR})$ which can be extracted from the respective proof, (BR) is the schema of (simultaneous) bar-recursion of Spector [95], here extended to the additional abstract types (see e.g. [40]) and $\mathcal{A}^\omega[X, \|\cdot\|]^-$ is the respective system without the axiom schemes QF-AC and DC.

This result extends to any suitable extension of the language of $\mathcal{A}^\omega[X, \|\cdot\|]$ (e.g. by any kind of new types and constants) together with any number of additional universal axioms in that (extended) language.

We now introduce the essential type and formula classes for the formulation of the metatheorem (essentially following [26, 39], see also [28]): We call a type ξ of degree n if $\xi \in T$ and it has degree $\leq n$ in the usual sense (see e.g. [40]). We call ξ *small* if it is of the form $\xi = \xi_0(0) \dots (0)$ for $\xi_0 \in \{0, X, X^*\}$ (including $0, X, X^*$) and call it *admissible* if it is of the form $\xi = \xi_0(\tau_k) \dots (\tau_1)$ where each τ_i is small and $\xi_0 \in \{0, X, X^*\}$ (also including $0, X, X^*$). Correspondingly, a formula F is called a $\forall$ -formula (respectively $\exists$ -formula) if $F = \forall \underline{a}^\xi F_{qf}(\underline{a})$ (respectively $F = \exists \underline{a}^\xi F_{qf}(\underline{a})$) with F_{qf} quantifier-free and all types ξ_i in $\underline{\xi} = (\xi_1, \dots, \xi_k)$ are admissible.

As mentioned before, besides Gödel's functional interpretation, the other central notion used in the bound extraction result is that of majorizability, going back to the fundamental work of Howard [30]. Besides of Howard's notion, we here also in particular rely on the subsequently extended notion of strong majorizability due to Bezem [9] which crucially, as shown by Bezem, provides a model of bar-recursion. The corresponding type structure of all strongly majorizable functionals, suitably extended to new abstract types, also forms the basis for the modern bound extraction theorems of proof mining as developed in [26, 39] and subsequent papers and thus also takes a central role here. We therefore discuss this model explicitly for our theory $\mathcal{B}^\omega$ based on the types T^{X, X^*} (essentially defined as in [77], but practically just derived from [26, 39]): At first, majorants of objects with types from T^{X, X^*} are objects with types from T according to the following projection:

Definition 5.4 ([77], essentially [26]). Define $\widehat{\tau} \in T$, given $\tau \in T^{X, X^*}$, by recursion on the structure via

$$\widehat{0} := 0, \widehat{X} := 0, \widehat{X^*} := 0, \widehat{\tau(\xi)} := \widehat{\tau}(\widehat{\xi}).$$

The majorizability relation for the types T^{X, X^*} and the accompanying structure $\mathcal{M}^{\omega, X, X^*}$ of all (strongly) majorizable functionals over a given normed space X with dual X^* is then defined recursively as follows:

Definition 5.5 ([77], essentially [26, 39]). Let $(X, \|\cdot\|)$ be a non-empty normed space with dual X^* . The structure $\mathcal{M}^{\omega, X, X^*}$ and the majorizability relation $\succsim_\rho$ are defined by

$$\left\{ \begin{array}{l} M_0 := \mathbb{N}, n \succsim_0 m := n \geq m \wedge n, m \in \mathbb{N}, \\ M_X := X, n \succsim_X x := n \geq \|x\| \wedge n \in M_0, x \in M_X, \\ M_{X^*} := X^*, n \succsim_{X^*} x^* := n \geq \|x^*\| \wedge n \in M_0, x^* \in M_{X^*}, \\ f \succsim_{\tau(\xi)} x := f \in M_{\widehat{\tau}}^{M_{\widehat{\xi}}} \wedge x \in M_\tau^{M_\xi} \\ \quad \wedge \forall g \in M_{\widehat{\xi}}, y \in M_\xi (g \succsim_\xi y \rightarrow fg \succsim_\tau xy) \\ \quad \wedge \forall g, y \in M_{\widehat{\xi}} (g \succsim_{\widehat{\xi}} y \rightarrow fg \succsim_{\widehat{\tau}} fy), \\ M_{\tau(\xi)} := \left\{ x \in M_\tau^{M_\xi} \mid \exists f \in M_{\widehat{\tau}}^{M_{\widehat{\xi}}} (f \succsim_{\tau(\xi)} x) \right\}. \end{array} \right.$$

Correspondingly, the full set-theoretic type structure $\mathcal{S}^{\omega, X, X^*}$ is defined via $S_0 := \mathbb{N}$, $S_X := X$, $S_{X^*} := X^*$ and

$$S_{\tau(\xi)} := S_\tau^{S_\xi}.$$

These structures later turn into models of the system $\mathcal{B}^\omega$ when they are equipped with suitable corresponding interpretations for the additional constants.

This, for modern proof mining essential and fundamental, idea of combining the Dialectica interpretation with the notion of majorizability goes back to [36] and was subsequently formally encapsulated through a novel type of functional interpretation, the so-called monotone functional interpretation, in [37]. In that way, also the present approach is thus “in spirit” of the monotone functional interpretation, where we write “in spirit” as we actually do not use a monotone variant of the Dialectica interpretation but treat the functional interpretation part and the majorization part of the combined interpretation separately, following the formal approach to proof mining metatheorems of [26, 39].

In any way, as is crucial for the applicability of this approach, many highly nontrivial mathematical statements nevertheless have a very simple monotone functional interpretation and one class of such principles is the so-called class Δ as already discussed in Section 2.2 before, which in particular accommodates all of the axioms of our system $\mathcal{B}^\omega$. Since we treat the majorization part separately from the functional interpretation part, we need a slightly modified approach to the treatment of statements of type Δ (where we essentially follow the approach given in [28], see also the recent [77]) and treat axioms of type Δ by employing a construction that converts a theory with axioms of such a type into a theory using only additional purely universal axioms formulated using the Skolem functions of these axioms. Concretely, we proceed as follows: Write $\widehat{\mathcal{B}}^\omega$ for $\mathcal{B}^\omega$ without any of its axioms of type Δ (which in particular covers all of the previous non-universal axioms $(*)_2$, $(*)_6$, $(f^*)_2$, $(f^{**})_2$) and without the rule (QF-LR) (which produces conclusions of type Δ). Given a set $\sqsupset$ of sentences of type Δ , we transform $\mathcal{B}^\omega + \sqsupset$ into a new theory $\overline{\mathcal{B}}^\omega_{\sqsupset}$ by adding to $\widehat{\mathcal{B}}^\omega$ the Skolem functionals $\underline{B}$ for any axiom of type Δ in $\mathcal{B}^\omega + \sqsupset$, say of the form

$$\forall \underline{a} \exists \underline{b} \leq_{\underline{\sigma}} \underline{r} \forall \underline{c} \leq_{\underline{\gamma}} F_{qf}(\underline{a}, \underline{b}, \underline{c})$$

where the types in $\underline{\delta}$, $\underline{\sigma}$ and $\underline{\gamma}$ are admissible and F_{qf} is quantifier-free, as new constants to the language and adding its ‘instantiated Skolem normal form’, i.e. the sentence

$$\underline{B} \leq_{\underline{\sigma}(\underline{\delta})} \underline{r} \wedge \forall \underline{a} \forall \underline{c} \leq_{\underline{\gamma}} F_{qf}(\underline{a}, \underline{B}\underline{a}, \underline{c})$$

as a new axiom. Further, we do the same with all conclusions of the rule (QF-LR): for any provable premise

$$\mathcal{B}^\omega + \sqsupset \vdash F_{qf} \rightarrow (\forall x^X, y^X, \alpha^1, \beta^1 (t(\alpha x +_X \beta y) =_{\mathbb{R}} \alpha t x + \beta t y) \wedge \forall x^X (|t x| \leq_{\mathbb{R}} M \|x\|_X))$$

with terms t and M , we add a new constant $x_{t,M}^*$ of type X^* to the language of $\overline{\mathcal{B}}^\omega_{\sqsupset}$ together with the corresponding axiom

$$\|x_{t,M}^*\|_{X^*} \leq_{\mathbb{R}} M \wedge (F_{qf} \rightarrow \forall x^X (t x =_{\mathbb{R}} \langle x, x_{t,M}^* \rangle_{X^*})).$$

Now, note that this new theory $\overline{\mathcal{B}}^\omega_{\sqsupset}$ extends $\mathcal{A}^\omega[X, \|\cdot\|]$ only by new types, constants and universal axioms. Therefore, Lemma 5.3 extends to this theory $\overline{\mathcal{B}}^\omega_{\sqsupset}$ where the conclusion is proved in $\overline{\mathcal{B}}^\omega_{\sqsupset} + (\text{BR})$ where $\overline{\mathcal{B}}^\omega_{\sqsupset}$ is again the same theory with the principles (QF-AC) and (DC) removed.

The main result on the majorizability part of the metatheorem is now the following model-theoretic lemma:

Lemma 5.6. *Let $\sqsupset$ be a set of additional sentences of type Δ . Let $(X, \|\cdot\|)$ be a (nontrivial) Banach space with its dual X^* . Let f be a convex, supercoercive and Fréchet differentiable function $f : X \rightarrow \mathbb{R}$ where ∇f , ∇f^* are uniformly continuous on bounded subsets and where*

f^* is supercoercive. Let A be a monotone operator $A : X \rightarrow 2^{X^*}$ with $A^{-1}0 \neq \emptyset$ such that the corresponding resolvents $\text{Res}_{\gamma A}^f$ are total for $\gamma > 0$.

Then $\mathcal{M}^{\omega, X, X^*}$ is a model of $\overline{\mathcal{B}}_{\Sigma}^{\omega-} + (\text{BR})$, provided $\mathcal{S}^{\omega, X, X^*} \models \sqsupset$ (with $\mathcal{M}^{\omega, X, X^*}$ and $\mathcal{S}^{\omega, X, X^*}$ defined via suitable interpretations of the additional constants). Moreover, for any closed term t of $\overline{\mathcal{B}}_{\Sigma}^{\omega-} + (\text{BR})$, one can construct a closed term t^* of $\mathcal{A}^{\omega} + (\text{BR})$ such that

$$\mathcal{M}^{\omega, X, X^*} \models \forall \omega^{0(0)(0)}, \omega'^{0(0)}, n^0 \left(\omega \gtrsim \omega^{\nabla f}, \omega^{\nabla f*} \wedge \omega' \gtrsim \alpha^f, \alpha^{f*} \right. \\ \left. \wedge n \geq_{\mathbb{R}} |f(0)|, \|\nabla f(0)\|_{X^*}, |f^*(0)|, \|\nabla f^*(0)\|_X, \|p_X\|_X \rightarrow t^*(\omega, \omega', n) \gtrsim t \right).$$

Proof. The proof is completely analogous to that of Lemma 7.7 in [77] (α^{f*} can be treated as α^f in [77]) after making the necessary additions for the treatment of A and its resolvents and zero which can be made similar as in the proof of Lemma 5.3 in [76]. But, for a more comprehensive and self-contained proof, we briefly detail these additions here: Concretely, we have to give interpretations of the additional constants χ_A , Res_A^f and p_X concerning A that were added to the system $\mathcal{D}_{f, f^*}^{\omega}[\text{FM}]$ to form $\mathcal{B}^{\omega}$ and then verify that these interpretations indeed have majorants to check that they belong to $\mathcal{M}^{\omega, X, X^*}$. The interpretations are immediately defined as follows (writing $\mathcal{M}$ as a shorthand for $\mathcal{M}^{\omega, X, X^*}$):

- (1) $[\chi_A]_{\mathcal{M}} := \lambda x \in X, x^* \in X^*. \begin{cases} 0 & \text{if } x^* \in Ax, \\ 1 & \text{otherwise,} \end{cases}$
- (2) $[\text{Res}_A^f]_{\mathcal{M}} := \lambda \alpha \in \mathbb{N}^{\mathbb{N}}, x \in X. \begin{cases} \text{Res}_{r_{\alpha} A}^f x & \text{if } r_{\alpha} > 0, \\ 0 & \text{otherwise,} \end{cases}$
- (3) $[p_X]_{\mathcal{M}} := p,$

where $p \in A^{-1}0$ is some zero of A as fixed in the formulation of the lemma. Take n, ω and ω' such that $\omega \gtrsim \omega^{\nabla f}, \omega^{\nabla f*}$ and $\omega' \gtrsim \alpha^f, \alpha^{f*}$ as well as $n \geq |f(0)|, \|\nabla f(0)\|, |f^*(0)|, \|\nabla f^*(0)\|, \|p\|$. Corresponding majorants for (1) and (3) are then easily given via

- (1)' $\lambda x^0, y^0. 1^0 \gtrsim [\chi_A]_{\mathcal{M}},$
- (3)' $n \gtrsim [p_X]_{\mathcal{M}}.$

We now focus on item (2). For this, take $b \gtrsim x$ as well as $\alpha \gtrsim \gamma$ (although the majorant will be independent of α here). By Lemma 3.12, we have

$$\left\| \text{Res}_{\gamma A}^f x \right\| \leq o(2D(b_n) + 2b_n C(b_n), b_n)$$

where $b_n = \max\{b, n\} + 1$ for the fixed n , where

$$o(\alpha, b) := F(\max\{\alpha^{f*}(\alpha + D(b) + b), 1\} + 1),$$

is defined as in Lemma 3.7 and

$$\begin{aligned} C(b) &:= b2^{\omega^{\nabla f}(0, b)} + [\|\nabla f(0)\|](0) + 2, \\ F(b) &:= b2^{\omega^{\nabla f*}(0, b)} + [\|\nabla f^*(0)\|](0) + 2, \\ D(b) &:= b2^{\omega^f(0, b)} + [|f(0)|](0) + 2, \end{aligned}$$

are defined as in Lemma 2.18. Since we have

$$C(b), F(b), D(b) \leq b2^{\omega(0, b)} + n + 3 =: E(b)$$

it immediately follows that

$$\begin{aligned} \left\| \text{Res}_{\gamma A}^f x \right\|_X &\leq o(2D(b_n) + 2bC(b_n), b_n) \\ &\leq E(\omega'(3E(b_n) + 2b_n E(b_n) + b_n) + 1) \\ &\leq E(\omega'(6b_n E(b_n)) + 1) \end{aligned}$$

and therefore we obtain

$$\lambda b^0.E(\omega'(6b_n E(b_n)) + 1) \gtrsim [\text{Res}_A^f]_{\mathcal{M}}. \quad \square$$

Theorem 5.7. *Let τ be admissible, δ be of degree 1 and s be a closed term of $\mathcal{B}^\omega$ of type $\sigma(\delta)$ for admissible σ . Let $\sqsupset$ be a set of sentences of type Δ . Let $F_\forall(x, y, z, u)/G_\exists(x, y, z, v)$ be $\forall$ -/ $\exists$ -formulas of $\mathcal{B}^\omega$ with only $x, y, z, u/x, y, z, v$ free. If*

$$\mathcal{B}^\omega + \sqsupset \vdash \forall x^\delta \forall y \leq_\sigma s(x) \forall z^\tau (\forall u^0 F_\forall(x, y, z, u) \rightarrow \exists v^0 G_\exists(x, y, z, v)),$$

then one can extract a partial functional $\Phi : \mathbb{N}^{\mathbb{N} \times \mathbb{N}} \times \mathbb{N}^\mathbb{N} \times \mathbb{N} \times S_\delta \times S_{\hat{\tau}} \rightarrow \mathbb{N}$ which is total and (bar-recursively) computable on $\mathbb{N}^{\mathbb{N} \times \mathbb{N}} \times \mathbb{N}^\mathbb{N} \times \mathbb{N} \times M_\delta \times M_{\hat{\tau}}$ such that for all $x \in S_\delta$, $z \in S_{\hat{\tau}}$, $z^ \in S_{\hat{\tau}}$ with $z^* \gtrsim z$ and for all $\omega \in \mathbb{N}^{\mathbb{N} \times \mathbb{N}}$, $\omega' \in \mathbb{N}^\mathbb{N}$, $n \in \mathbb{N}$ with $\omega \gtrsim \omega^{\nabla f}, \omega^{\nabla f^*}$, $\omega' \gtrsim \alpha^f, \alpha^{f^*}$ and $n \geq |f(0)|, \|\nabla f(0)\|, |f^*(0)|, \|\nabla f^*(0)\|, \|p\|$:*

$$\begin{aligned} \mathcal{S}^{\omega, X, X^*} \models \forall y \leq_\sigma s(x) (\forall u \leq_0 \Phi(\omega, \omega', n, x, z^*) F_\forall(x, y, z, u) \\ \rightarrow \exists v \leq_0 \Phi(\omega, \omega', n, x, z^*) G_\exists(x, y, z, v)) \end{aligned}$$

holds whenever $\mathcal{S}^{\omega, X, X^} \models \sqsupset$ for $\mathcal{S}^{\omega, X, X^*}$ defined (via a suitable interpretation of the additional constants similar to [77] and [76], recall Lemma 5.6) using any (nontrivial) reflexive Banach space $(X, \|\cdot\|)$ with its dual X^* and using a convex, supercoercive (with modulus α^f) and Fréchet differentiable function $f : X \rightarrow \mathbb{R}$ where $\nabla f, \nabla f^*$ are uniformly continuous on bounded subsets (with moduli $\omega^{\nabla f}, \omega^{\nabla f^*}$, respectively) and where f^* is supercoercive (with modulus α^{f^*}) and using a monotone operator $A : X \rightarrow 2^{X^*}$ with $p \in A^{-1}0 \neq \emptyset$ such that the corresponding resolvents $\text{Res}_{\gamma A}^f$ are all total for $\gamma > 0$.*

Further:

- (1) *If $\hat{\tau}$ is of degree 1, then Φ is a total computable functional.*
- (2) *We may have tuples instead of single variables x, y, z, u, v and a finite conjunction instead of a single premise $\forall u^0 F_\forall(x, y, z, u)$.*
- (3) *If the claim is proved without DC, then τ may be arbitrary and Φ will be a total functional on $\mathbb{N}^{\mathbb{N} \times \mathbb{N}} \times \mathbb{N}^\mathbb{N} \times \mathbb{N} \times S_\delta \times S_{\hat{\tau}}$ which is primitive recursive in the sense of Gödel [27] and Hilbert [29]. In that case, also plain majorization can be used instead of strong majorization.*

Proof. The proof now is completely analogous to that of Theorem 7.8 in [77] and we just briefly sketch it: Assuming the premise, one first realizes that the same statement is also provable in $\bar{\mathcal{B}}_\sqsupset^\omega$. Using the functional interpretation, i.e. Lemma 5.3, for this system, one extracts realizers for (the $\forall\exists$ -prenexiation of) this statement which are then majorized in the model based on $\mathcal{M}^{\omega, X, X^*}$ through Lemma 5.6. Utilizing that the types in the statement are admissible, one can then in turn conclude that the resulting bounds are also valid in the model based on $\mathcal{S}^{\omega, X, X^*}$. The additional statements (1) – (3) can also be concluded similarly as in [77]. $\square$

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